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If the location of firm operations is relevant for financing, multinationals should have easier access to different foreign sources of funding relative to domestic firms. I document that U.S. multinationals are more likely to borrow from a foreign bank and to issue international bonds than are U.S. domestic firms. Multinationals are less affected than domestic firms by capital market dislocations because of greater funding flexibility. Using the 2007–2009 financial crisis as a capital supply shock, I find that multinationals relied more on foreign funding sources in bank loans and consequently reduced domestic investment less than did domestic firms. (*JEL* G01, G15, G31, G32)

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The operations of U.S. firms are largely internationally diversified; about 54% of the publicly traded U.S. firms have operations in foreign countries, and these activities represent 35% of their net income.[1] One of the benefits of international corporate diversification, often cited by managers and textbooks, is that expanding operations overseas can improve access to capital markets and lower the cost of capital.[2] In this paper, using a sample of U.S. multinational

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[1] The statistics are as of the fiscal year-end 2010 using a sample of U.S. headquartered public companies in the Compustat database.

[2] International business textbooks often rationalize international corporate diversification because it improves access to capital markets (Hill 2005; Shapiro 2008). For example, Chinese banks are recently increasing their lending in U.S. syndicated loan markets as U.S. companies diversify their funding sources. Dell Inc., one of the large U.S. multinational firms, said that it had developed relationships with Chinese banks, which made it easier to conduct business in China (Wong 2012).

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and domestic firms, I evaluate the way in which international activities affect the debt financing policies of firms.[3]

If global capital markets are not perfectly integrated across countries, foreign funding is potentially more costly to obtain than domestic funding. The extent to which borders increase the cost of finance and to which having operations in different countries limits this increase is an important and unresolved question. If having an operation presence in foreign countries reduces frictions in international debt markets, multinational firms will have easier access to different foreign sources of funding than domestic firms. Consequently, since multinational firms can access international capital markets at a lower cost, they should have greater financing flexibility than domestic firms. A particularly important implication of this flexibility is that U.S. multinational firms' financing, and hence their investments, should be less affected by capital market disruptions in the United States than otherwise similar domestic firms.

An example of the way in which the financing sources of multinationals can differ from those of domestic firms is the case of Fuel System Solutions. Its loan financing sources are diversified across countries, as the firm receives bank loans from multiple lenders in the United States, Brazil, and Italy through its foreign subsidiaries. For example, in December 2004, its Italian subsidiary entered into a five-year unsecured term loan of $13.6 million from an Italian bank, Unicredit Banca Medio Credito S.p.A. Maintaining foreign funding sources was beneficial especially during the 2007–2009 financial crisis. Fuel Systems Solutions was not able to renew the existing U.S. credit facility from LaSalle Business Credit, which matured in January 2008. However, in December 2008, the firm expanded its business through an acquisition, and the costs were funded through a 1.5 year term loan of $6.7 million from two Italian banks (Banca IMI S.p.A and Intesa San Paolo, S.p.A). This anecdotal evidence illustrates the financial benefit of international corporate diversification. While the firm experienced credit constraints in the United States lending markets during the financial crisis, it was still able to obtain funding through its foreign subsidiaries from local lenders in Europe and Latin America, which were relatively less affected by the crisis.

In this paper, I empirically test the prediction that multinational firms are more financially flexible than purely domestic firms by maintaining multiple sources of capital across countries. To do so, I use a sample of bank loans and corporate bonds issues between 2000 and 2010 to compare capital raising activities of U.S. multinationals and domestic firms. However, establishing a causal relation is empirically challenging because multinationals are likely to be systematically different from domestic firms. On average, multinationals are larger and have a lower volatility of cash flows than domestic firms, both of which reduce multinationals' credit risk and cost of funding. These underlying

[3] Throughout the paper, I use the terms "global diversification," "international corporate diversification," and "being multinational" interchangeably.

differences, as opposed to the location of their assets, could possibly explain differences in access to capital.

To isolate the effect of being multinational given these differences, I use three identification strategies. First, I employ propensity-score-matching methods to match loans and bonds issued by multinationals to comparable loans and bonds issued in the same quarter by domestic firms with similar observable characteristics. Second, I exploit the geographic heterogeneity in multinationals' operations at a country level to identify the mechanism through which multinationals can achieve better access to capital. Using comprehensive data on U.S. firms' subsidiary structure from Capital IQ, I examine the impact of a subsidiary's location effect on the use of foreign funding within multinational firms. Data on the location of multinationals' foreign subsidiaries allow me to evaluate whether differences in financing patterns occur merely because multinationals tend to be larger and less risky firms. Third, I use the 2007–2009 financial crisis as a capital supply shock, during which having multiple sources of funding is particularly valuable. Results from these three approaches suggest that it is the multinational status of the firms, rather than other attributes of these firms, that leads to multinationals' improved access to capital.

I first document that because they are internationally diversified, multinationals have easier access to financing from foreign countries than purely domestic firms, which leads to more diversification in funding sources. Using a sample of 6,455 bank loans issued by multinationals and 5,184 loans by domestic firms, I estimate whether a loan is originated by domestic or foreign lenders. The results indicate that the probability a multinational firm has at least one foreign lender in its bank loan syndicate is 13.3 percentage points higher than for a domestic firm, which is a 32% increase relative to domestic firms. Multinationals are also more likely than domestic firms to issue corporate bonds abroad. Using a sample of 2,209 corporate bonds issued by multinationals and 1,197 bonds issued by domestic firms, I find that multinationals are 5.2 percentage points more likely than domestic firms to place a corporate bond in international capital markets, which is 23% greater than the predicted likelihood for comparable domestic firms.

If multinational firms are able to access foreign funding in ways that purely domestic firms cannot, multinationals should have greater financial flexibility when faced with credit supply shocks in their home countries. In particular, I expect multinationals to be less financially constrained than domestic firms if there are credit supply shocks in their home countries. Using the 2007–2009 financial crisis as a supply shock to capital, which hit the U.S. credit market more severely than some other foreign markets, I compare the differences in financing and investment policies between U.S. multinationals and domestic firms before and after the crisis. The results suggest that the probability of receiving bank loans from foreign lead lenders significantly increased for multinationals at the peak of the crisis (the two quarters after the Lehman bankruptcy) relative to the period outside the financial crisis, but the probability of receiving bank loans did not change for domestic firms. Furthermore, the propensity to issue an international bond sharply increased by 23.7 percentage points more for multinationals than for domestic firms in the two quarters following September 2008 (Lehman bankruptcy) relative to the noncrisis period.

Of the loans and bonds issued by multinationals, the increased likelihood of foreign loan issuance during the crisis is mostly driven by the cases in which lead lenders are from the countries in which multinationals have foreign subsidiaries. In addition, the effect is more pronounced for multinational firms with subsidiaries in the countries in which private bank lending was not severely affected during the crisis. These cross-sectional analyses using the multinationals' subsidiary structure suggest that the existence of foreign operations, rather than other attributes of multinational firms, is what enables multinational firms to access international capital markets when domestic credit is scarce.

I then explore whether having access to foreign capital markets affects firms' investment. If alternative sources of foreign funding mitigate U.S. credit supply contractions, multinationals' investments should be less affected by U.S. credit market conditions than domestic firms' investments. To control for the different demand shocks in domestic and foreign markets during the crisis, I compare U.S. capital expenditures and U.S. expansion announcements of multinational firms to those of domestic firms. The main finding is that multinational status is positively associated with firms' domestic investment during the crisis period. Domestic firms' ratio of capital expenditures to total assets decreased by between 1.8 and 2.1 percentage points in 2009 and in 2010, yet multinationals did not experience a significant change in U.S. capital expenditures during that period. This result holds regardless of the way in which I control for the investment opportunities and the cash flows in the domestic operations. In particular, multinationals' U.S. investment was not adversely affected when they were able to issue foreign debt during the crisis period.

The different responses in financing and investment to the capital supply shock are consistent with the prediction that multinationals are better able to exploit alternative funding sources outside the United States when they are credit constrained in the United States. Partly because of their greater financial flexibility, multinational firms did not cut their U.S. investment as much as domestic firms did during the crisis.

The evidence of the financial flexibility of multinationals in this paper is related to the debate regarding the effect of capital supply on firms' financial and investment policies. As supported in previous empirical studies, the extent to which the supply of capital affects firms' financial and investment policies varies, depending on the financial constraints of borrowers and the tightness of credit conditions (see Faulkender and Petersen 2006; Leary 2009; Lemmon and Roberts 2010; Erel et al. 2012). The implication of my paper is that a firm's geographical business structure is one of the determinants of access to capital, and this implication is in addition to factors recognized in the literature, which include firm size, firm leverage, and whether a firm has a bond rating. As a consequence, the investment of multinationals would be financed from both U.S. and foreign sources of capital, which puts these multinationals at a relative advantage to domestic firms that mostly rely on domestic funding when the supply of domestic capital is low.

My study also contributes to the large literature on the way in which the organizational structure of firms affects firm value. Focusing on the external capital markets, this paper emphasizes an additional financial benefit of diversifying operations abroad: the option to diversify sources of funding across countries. While theoretical models have been developed to explain the financial benefit of corporate diversification (Stein 1997; Scharfstein and Stein 2000; Rajan, Servaes, and Zingales 2000), the empirical evidence, which mainly focuses on industrial diversification, is still the subject of debate. Applying the same argument to geographic diversification, several papers compare the impact of both industrial and international diversification on overall valuation (Denis, Denis, and Yost 2002; Bodnar, Tang, and Weintrop 2003; Creal et al. 2014). By closely examining capital raising activities and investments of U.S. multinationals and domestic firms, I provide evidence of a specific channel through which international corporate diversification creates value in alleviating financial constraints in global capital markets.[4] This result supports the recent finding that the value of globally diversified firms increased during the financial crisis (Chang, Kogut, and Yang 2016).

Lastly, this study also extends a literature on bank lending in an international setting. A large banking literature has documented that the geographic proximity of a firm with its lender is correlated with the effective monitoring and the existence of lending relationships (Petersen and Rajan 2002; Brickley, Linck, and Smith 2003; Degryse and Ongena 2005; Agarwal and Hauswald 2010; Arena and Dewally 2012). However, those studies focus on the local effects mostly within a country or a region. Given that financial frictions are more severe and enforcement of debt contracts is more difficult across international borders than within a country, the location of assets would significantly matter for global lending decisions. My study highlights the importance of the firms' international presence on international debt financing decisions, especially for bank loans that require soft information and frequent monitoring. On the other hand, the results on corporate bond markets in this study also add to the previous studies that have explored how firms use international and domestic bond markets differently (Gozzi et al. 2013; Massa and Žaldokas 2014). The relevant question following these studies is "which firms can take advantage of both markets?" My paper provides some evidence that helps address this

[4] This evidence also supports the findings in early studies in international finance that investors positively value multinationality (see Errunza and Senbet 1981, 1984; Morck and Yeung 1991). In addition, it adds new evidence to the previous studies that investigate the capital structure and the internal capital markets of multinationals (e.g., Desai, Foley, and Hines 2004; Desai, Foley, and Forbes 2008) by showing how multinational firms use the external capital markets.

question; having foreign assets allows firms to exploit a broader set of both domestic and international investors in corporate debt markets.

1. Data and Descriptive Statistics

1.1 Data and variables

To construct my sample, I use the Compustat Fundamentals Quarterly database to identify all publicly traded firms incorporated and headquartered in the United States between 2000 and 2010.[5] I exclude financials and utilities (SIC codes 4900–4949 and 6000–6999) and further restrict the sample to firms with firm-quarter observations with sales greater than $20 million, positive total assets, positive cash and marketable securities, and cash and marketable securities less than total assets. The panel consists of 130,902 firm-quarter observations and 6,203 firms.

A firm is defined as a multinational if any of its foreign pretax income (Compustat item: PIFO) or foreign income tax (Compustat item: TXFO) is not missing in at least one year over the previous three years, and the firm must have at least one subsidiary outside the United States. To identify firms that have foreign operations, I utilize information on income tax expenses reported in the annual financial statements. If pretax income or deferred taxes for non-U.S. operations exceed 5% of the consolidated total, the SEC (SEC Regulation §210.4-08(h)) requires firms to disclose pretax income and deferred taxes for U.S. and non-U.S. operations separately. Using information on the existence of foreign subsidiaries, I confirm that those multinationals actually have physical assets outside the United States.[6] To determine the existence and locations of foreign subsidiaries, I use the Capital IQ database for subsidiary structures, which is mainly sourced from SEC regulatory filings.[7] Both direct subsidiaries and indirect subsidiaries owned by direct subsidiaries are included, but subsidiaries in tax-haven countries are not counted.[8] Based on the definition

5 To construct historical subsidiary structures, I also use the Capital IQ transaction database (e.g., spin-offs, mergers, and acquisitions) so that any changes in the subsidiary-parent relationship caused by those transactions are taken into account. The sample ranges from 2000 to 2010 because this transaction database is available from 1998.

6 Some papers use information on foreign sales from the Compustat Geographic Segment database to define internationally diversified firms (see, e.g., Denis, Denis, and Yost 2002). According to this definition, however, a firm that exports goods to other countries would be defined as a multinational firm even if it does not have any assets outside the United States. Using foreign pretax income is a better way to define a multinational for the purpose of this paper, because this approach excludes firms that do not have foreign operations, hence any income source from other countries. Moreover, according to SFAS no. 14, a firm is required to disclose segment-level data on sales, income, or assets from operations outside the United States if they account for more than 10% of its consolidated value. As the threshold of 5% for foreign pretax income report regulation is lower than that of 10% for the geographical segment data, the measure based on the foreign pretax income allows me to identify a broader set of firms with foreign operations.

7 Any size-related information, such as sales and assets at the subsidiary level, is incomplete in Capital IQ. Thus, I only use information on the names and locations of subsidiaries across countries.

8 I exclude tax-haven subsidiaries because it is likely that those subsidiaries do not have material operations beyond avoiding or lowering taxes (Desai, Foley, and Hines 2006). The list of tax-haven countries is obtained from OECD (2004).

of multinational and domestic firms, I construct the main variable, *GlobalDiv*, which takes a value of one in a given year if the firm is multinational and zero otherwise. The final sample includes 2,353 multinationals and 3,850 domestic firms.

To examine the extent to which firms can access bank loan markets, I use DealScan to construct the data on syndicated and sole lender bank loans issued by the list of the firms identified above or by their subsidiaries from 2000 to 2010.[9] I use the loan facility as the unit of analysis, for the borrower-lender relationship is facility specific.[10] The dependent variables in the bank loan analysis are the probability of foreign lender participation and the share of the loan retained by foreign lenders. Since lead arrangers usually make loan contract decisions and monitor, I also consider lead lender participation separately. I classify lenders incorporated outside of the United States as foreign lenders based on the information on the nationality of each lender provided by DealScan.[11]

For corporate bond issuance, I include public and privately placed corporate bonds issued by the sample firms or by their subsidiaries in the 2000 to 2010 period from SDC, but I exclude those issued by the financial subsidiaries.[12] Unlike bank loans, multiple tranches of one bond issue are placed in the same market, so I aggregate observations with multiple tranches at a bond level by summing the proceeds and setting the maturity equal to the proceeds-weighted average. As a primary variable of interest in corporate bond analysis, I consider whether the bond is issued in international markets. I classify a bond as an international one if it is placed in exchanges outside the United States, if it is a Euro bond, or if it is a global bond.

From the bank loan and corporate bond samples, I exclude non-US-dollar-denominated loans and bonds. Multinationals have an incentive to receive debt denominated in a foreign currency to hedge the foreign currency risk of their revenues from foreign operations, but domestic firms do not share this incentive. Debt financing from international markets may be more available in the form of foreign currency debt. To the extent that multinationals have a natural demand for foreign currency-denominated debt, it is possible for them to access foreign

9 Not all subsidiary debt is covered by DealScan or SDC, particularly if the amount of bank loans is not substantial or if they are not syndicated. In this sense, the debt issuance data in my analysis potentially underestimate the actual magnitude of multinational's access to foreign capital markets.

10 Each loan reported in DealScan contains one or multiple facilities. The final sample includes 11,639 loan facilities associated with 9,482 deals. Within the same loan deal, each loan facility can have different levels of lender participation depending on the types of facilities. Revolving credit facility and a term loan A are typically held by banks, whereas a term loan B is funded by institutional investors (e.g., Ivashina and Sun 2011). Analysis with the data aggregated at the deal level does not influence the main results of this paper.

11 A foreign branch of U.S. banks or a U.S. branch of foreign banks is not classified as a foreign lender. Such an approach is extremely conservative, for it treats U.S. branches of foreign banks as U.S. banks.

12 If the SIC code of a subsidiary issuer is between 6000–6999 or its first two digits of NAICS is equal to 52, then the subsidiary issuer is defined as a financial arm of subsidiary (e.g., GE Capital Australia).

currency borrowing at lower cost. Therefore, restricting the sample to US-dollar-denominated loans allows me to control for the demand of multinational firms for foreign borrowing for hedging purposes.[13]

Finally, the bank loan and corporate bond issuance data sets are merged with the quarterly firm-level accounting data from Compustat for the quarter prior to the issue dates. The final bank loan issuance data consist of 11,639 loan facilities, out of which 6,455 loans are made to 1,379 multinationals and 5,184 loans are made to 1,459 domestic firms. The final corporate bond issuance data has 3,406 bond issuances in total, 2,209 of which are issued by 602 multinationals and 1,197 of which by 367 domestic firms.

To measure domestic investment of multinationals and domestic firms, I consider capital expenditure announcements and expansion announcements. *U.S. capital expenditure* is defined as the annual capital expenditures of U.S. operations scaled by the total assets. I obtain the data on the annual U.S. segment capital expenditures for multinational firms from the Compustat Geographical segment database, and I use the aggregate capital expenditures for domestic firms. SEC's segment reporting rule (SFAS No. 131) requires a firm to disclose geographical segment-level revenues and assets, but other items (e.g. capital expenditures, operating profits) are only voluntarily disclosed. Because of the limited availability of segment-level capital expenditure information, 14% of firm-year observations of multinational firms will be used. To supplement the segment-level capital expenditure data, I also construct *U.S. expansion announcement*, which takes a value of one for a quarter in which a firm makes either U.S. acquisition announcements or U.S. expansion announcements. For each sample firm, I identify the U.S. acquisition announcements from SDC, which include completed transactions of acquiring U.S. assets of other firms or partial/entire equity stakes on U.S. target firms by the sample firms. Expansion announcements from Capital IQ include "Seeking Acquisitions/Investments" and "Business Expansion" events from the "Key Development" database, and then I identify corporate news regarding the expansions in the U.S. operations by reading individual news articles.[14]

Following previous studies on the 2007–2009 financial crisis (e.g., Kahle and Stulz 2013), I divide the financial crisis period into *Early crisis,* which is defined as calendar quarters between 2007Q3-2008Q3 and *Late crisis* as 2008Q4-2009Q1, which is after Lehman collapsed. It was not until the fourth quarter of 2008 that the credit spreads in corporate lending markets increased substantially. Hence, having two crisis period indicators allows me to capture

13 In untabulated results, I find that the main results of the paper become stronger when I include non-US-dollar-denominated loans and bonds in the sample. In this sense, the results from the sample of bank loans and corporate bonds restricted to the US-dollar-denominated debt may underestimate the actual magnitude of the financial benefit of international diversification.

14 Examples of the corporate news headlines are as follows: "Convergys Corp. to Open New Facilities in Kentucky and Texas," which is identified as an U.S. expansion announcement, and "Kraft to Invest into a New Manufacturing Line in Russia," which is identified as a foreign expansion announcement.

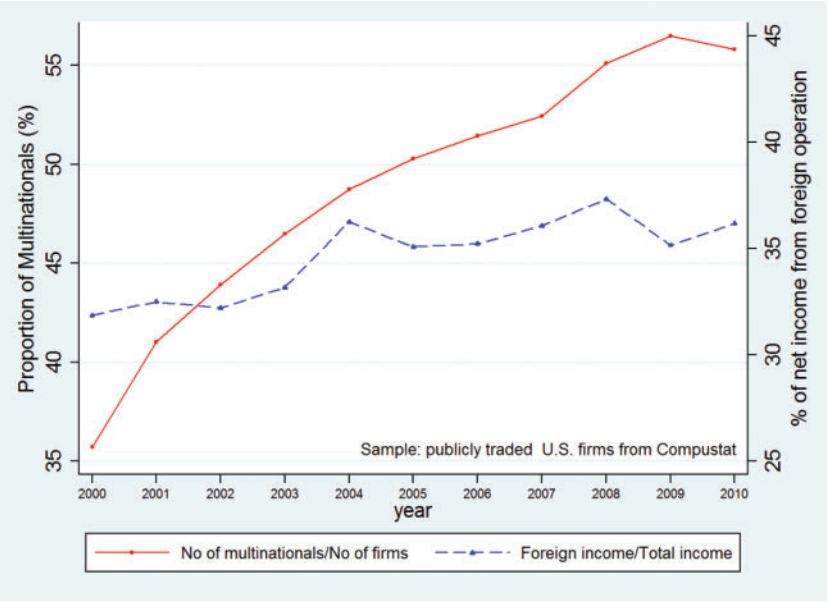

Figure 1
The proportion of multinationals and foreign income
This figure plots the proportion of multinationals and the average percentage of foreign income of U.S. firms from 2000 to 2010. *Multinational* is defined as a firm that does not have missing foreign pretax income for at least one year over the previous three years and has at least one subsidiary in countries outside the United States. Otherwise, a firm is defined as *Domestic*. The percentage of foreign income is defined as abs(foreign income)/[abs(domestic income) + abs(foreign income)].

the different effect of international corporate diversification based on the timing of the crisis. As alternative crisis measures, I also confirm the main results with three continuous variables that capture the intensity of the crisis: the *VIX index* (Chicago Board Options Exchange Volatility Index), the *TED spread* (the three-month LIBOR over Treasury-bill yields), and the *CP spread* (the three-month commercial paper rate over Treasury-bill yields).

To control for other factors that would affect financial and investment policies of the firms, I include firm, loan, and bond-level characteristics in multivariate regressions. All variables are winsorized at 1% and 99%, and Table A1 (see the appendix) presents definitions for all the variables.

1.2 Summary statistics

Figure 1 shows the evolution of the fraction of multinationals and changes in foreign profits over time. About 35.5% of firms have foreign operations at the end of 2000, and the proportion of multinationals is constantly increasing through the sample period to over 55% in 2010. In panel A of Table 1, I look at the distribution of multinationals based on the intensity of their foreign operations. On average, multinationals generate more than 32% of their total income and

Table 1
Descriptive statistics of multinational's foreign operations and difference in firm characteristics between multinationals and domestic firms

A. Intensity of foreign operations and location of foreign subsidiaries of multinationals

Foreign operation involvement	Mean	Median	SD
% foreign income	34.8	27.9	28.4
% foreign sales	32.7	29.4	24.2
HHI(sales)	0.473	0.490	0.268
Total number of subsidiaries	41.475	20.000	63.677
Number of foreign subsidiaries	20.638	7.000	39.784
Number of countries in which MNCs have foreign subsidiaries	10.783	6.000	12.286
Observations	15,520		

Location of subsidiaries

% by region		% by country (top 10)	
Europe	81.7	United Kingdom	63.2
Asia	66.0	Canada	57.7
Latin America	40.9	Germany	45.5
Middle East	16.0	France	39.8
Africa	15.7	Netherlands	36.8
		Australia	34.9
		China	33.0
		Japan	32.1
		Singapore	30.1
		Mexico	29.9
Observations	15,520		

B. Firm characteristics

	Mean			Median		
Variables	Multinational (1)	Domestic (2)	Difference (1)-(2)	Multinational (3)	Domestic (4)	Difference (3)-(4)
Total assets ($ billion)	5.100	1.238	3.862***	0.750	0.194	0.556***
Market cap ($ billion)	5.892	1.110	4.782***	0.782	0.189	0.593***
Log(sales)	5.248	4.010	1.238***	5.170	3.820	1.350***
Leverage	0.208	0.230	−0.022***	0.180	0.172	0.008***
Sales growth	0.110	0.168	−0.059***	0.071	0.077	−0.007***
Cash flows	0.030	0.022	0.008***	0.030	0.027	0.003***
Cash	0.179	0.185	−0.006	0.108	0.088	0.020***
Market-to-book	1.928	1.921	0.006	1.538	1.420	0.118***
STD(cash flows)	0.018	0.026	−0.008***	0.013	0.018	−0.005***
Capex	0.011	0.016	−0.004***	0.007	0.008	−0.001***
U.S. capital expenditure	0.045	0.065	−0.020***	0.024	0.036	−0.012***
U.S. expansion announcement	0.204	0.126	0.078***			
Not rated	0.613	0.801	−0.188***			
Observations	62,885	68,017		62,885	68,017	
Number of firms	2,353	3,850		2,353	3,850	

This table provides descriptive information on the U.S. multinationals and domestic firms in the sample. The sample includes firm-quarter observations of all publicly traded U.S. firms from 2000Q1 to 2010Q4 from the Compustat database, excluding financials and utilities (SIC codes 4900–4949 and 6000–6999). Firm-quarter observations with annual sales less than $20 million, negative total assets, negative cash and marketable securities, and cash and marketable securities greater than total assets are deleted. A firm is defined as *Multinational* if its foreign pretax income is not missing in at least one year over the previous three years and there is at least one subsidiary in a country outside the United States. Otherwise, a firm is defined as *Domestic*. Panel A describes the structure of foreign operations of 2,353 multinationals. Both direct subsidiaries and indirect subsidiaries owned by direct subsidiaries are included, but subsidiaries in tax-haven countries are not counted. Data on foreign operation structure are on annual basis. Panel B presents the summary statistics of firm characteristics of multinationals and domestic firms using firm-quarter observations from 2000Q1 to 2010Q4. The tests of differences in means (medians) between multinationals and domestic firms are based on univariate OLS (median) regressions, where each variable is regressed on the indicator of multinational. Table A1 (see the appendix) defines the variables. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

sales overseas and have subsidiaries in ten countries on average (six countries in median). More than half of the multinationals have subsidiaries in the United Kingdom or Canada, and more than 30% of multinationals have operations in one of the Asian countries such as China, Japan, and Singapore. These statistics imply that even within multinationals, there is substantial heterogeneity in the intensity of foreign operations and the location of foreign subsidiaries.

In panel B, I report the summary statistics of firm characteristics of multinationals and domestic firms and I find a considerable difference between the two groups. Multinationals are significantly larger in terms of both market values and total assets. The mean (median) total asset size of multinationals is $5.10 billion ($0.75 billion), while that of domestic firms is $1.24 billion ($0.19 billion). Multinationals are also more profitable than domestic firms in terms of cash flows, but they have relatively lower sales growth and leverage than domestic firms.[15] Multinationals are less risky firms than domestic firms, for they have lower cash flow volatility and are more likely to be rated. The domestic U.S. capital expenditures of multinationals are lower than those of domestic firms. These substantial differences between multinationals and domestic firms emphasize the importance of considering the possibility that other sources of heterogeneity could affect access to foreign capital. For this reason, I employ a propensity-score-matching method in Sections 2 and 3.

2. International Corporate Diversification and Access to Foreign Capital

2.1 Baseline regressions

I first consider the way in which firms' geographic structure affects the use of foreign funding in bank loans and corporate bonds. Panel A of Table 2 displays summary statistics for the sample of the bank loan facilities. Conditional on having bank loans, multinationals rely more heavily on foreign lenders than domestic firms, both in terms of number and volume. Although 42.2% of the loans issued by domestic firms have more than one foreign lender, that figure is significantly lower than the 65.7% of the loans issued by multinationals. In addition, foreign lenders retain 22.1% of the total facility amount for loans to multinational borrowers, which is in contrast to 12.2% for loans to domestic borrowers. The difference between multinational and domestic firms, in terms of whether a lead arranger of the loan facility is a foreign bank, is also substantial; the percentage of loans from foreign lead lenders is 17% for multinationals, as opposed to 10.8% for domestic firms. In addition, a large portion of foreign lender shares of multinational firms is retained by lenders from countries in which they have foreign subsidiaries (on average 13.1%).

[15] The fact that multinationals have lower leverage is consistent with the finding of previous studies of international diversification (e.g., Lee and Kwok 1988; Doukas and Pantzalis 2003). However, the main focus in this paper is not on the difference of absolute level of leverage between multinationals and domestic firms, but on funding sources and changes in use of foreign funding over the crisis period.

Table 2
Summary statistics of bank loan and corporate bond issuance

A. Bank loans

	Lender			Lead lender		
	Multinational (1)	Domestic (2)	Difference (1)-(2)	Multinational (3)	Domestic (4)	Difference (3)-(4)
Lender participation						
Number of lenders	9.412	6.652	2.759***	6.652	1.336	0.297***
Lead lender shares				0.353	0.457	−0.104***
Have foreign lenders	0.657	0.422	0.235***	0.170	0.108	0.063***
Foreign lender shares	0.221	0.122	0.099***	0.043	0.029	0.013***
Loan characteristics						
Facility amount ($MM)	508.331	244.660	263.672***			
Maturity (month)	44.431	46.347	−1.916***			
Secured	0.469	0.652	−0.183***			
Missing secured	0.307	0.213	0.094***			
Revolver	0.568	0.615	−0.047***			
Term loan	0.256	0.301	−0.046***			
Observations	6,455	5,184	11,639			
Number of firms	1,379	1,459				

B. Corporate bonds

	Multinational (1)	Domestic (2)	Difference (1)-(2)
Bond placement			
International bond	0.368	0.229	0.139***
Bond characteristics			
Proceed amount ($MM)	639.272	422.531	216.741***
Maturity (month)	111.588	117.166	−5.578
Secured	0.034	0.075	−0.041***
Private debt	0.312	0.497	−0.185***
Callable	0.248	0.398	−0.150***
Observations	2,209	1,197	3,406
Number of firms	602	367	

This table reports summary statistics of bank loan issuances and corporate bonds by multinationals and domestic firms. In panel A, data include all US-dollar-denominated syndicated loans or sole-lender bank loans issued by publicly traded U.S. firms during the 2000 to 2010 period from DealScan. The sample also includes loans issued by subsidiaries whose parent firms are publicly traded U.S. firms. Loan facility is used as the unit of analysis. *A foreign lender* is defined as a lender incorporated outside of the United States; this excludes a foreign branch of U.S. banks or a U.S. branch of foreign banks. A lender is defined as *a lead lender* if "Lead Arranger Credit" is equal to "Yes" in DealScan, if the loan is a sold-lender loan, or if a lender is identified as "Agent," "Administrative Agent," "Arranger," or "Lead Bank." For loans without share data, I assume each lender in the loan facility has an equal share. In panel B, data include all US-dollar-denominated public or private bonds issued by U.S. publicly traded firms during the 2000 to 2010 period from SDC. The sample also includes bonds issued by subsidiaries whose parent firms are publicly traded U.S. firms. A bond is defined as *an international bond* if the bond is placed in an exchange market outside the United States, if it is a Euro bond, or if it is a global bond. The tests of differences in means between multinationals and domestic firms are based on univariate OLS regressions in which each variable is regressed on the indicator of multinational. Table A1 (see the appendix) defines the variables. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

In addition, I also find that multinationals access international bond markets more frequently than domestic firms. While only 22.9% of bonds issued by domestic firms are international bonds, 36.8% of bonds issued by multinationals are issued in international market, and the difference between these percentages is statistically significant. The summary statistics also suggest that multinational and domestic firms tend to issue loans and bonds with different features. For example, debt issued by multinationals is more likely to be larger and less secured than those issued by domestic firms. These differences in loan and bond features are controlled in the multivariate regression analysis below.

To evaluate whether firms' multinational status affects the probability of foreign lenders' participation and the fraction of loan amount from foreign lenders in bank loan markets, I estimate a probit model predicting whether a loan facility's syndicate includes at least one foreign lender. Separately, I also consider whether a lead lender is a foreign bank or not. The main independent variable is *GlobalDiv*, and control variables include firm-specific characteristics and loan facility features.[16] Firm-level controls include *Log(sale), Sales growth, Cash flows, Market-to-book, Leverage, Cash,* and *STD(cash flows)*. The regression also includes S&P credit rating fixed effects and industry fixed effects at the one-digit SIC industry level. Loan characteristics include *Log(facility amount), Log(maturity), Secured, Missing secured,* and *Log(number of lenders)*, as well as deal purpose and loan facility type fixed effects. Standard errors are clustered at the firm level. A probit model of foreign lenders' participation does not consider the importance of foreign lenders' role in terms of volume. To examine the magnitude of foreign lender participation, I also estimate an ordinary least-squares (OLS) regression model in which the dependent variables are foreign lender shares and foreign lead lender shares.

Panel A of Table 3 reports the marginal effects from probit regressions in Columns 1 and 2 and the coefficients from OLS regressions in Columns 3 and 4. The results indicate a significantly positive relation between the likelihood of foreign lender participation and the borrower's multinational status. Being multinational increases the probability of having a foreign lender in the loan facility syndicate by 13.3 percentage points, which is about a 32% increase, given that 42.2% of loans to domestic firms have at least one foreign lender in the loan facility syndicate. When I focus on the lead lenders in Columns 2 and 4, the effect is also statistically significant; being multinational increases the probability of borrowing from a non-U.S. lead bank by 4%, which is equivalent to a 38% increase compared to the proportion of loans to domestic firms from foreign lead lenders. The results in Columns 3 and 4 confirm a significantly positive relation between the foreign lender participation and being a multinational firm in terms of the volume.

In addition to bank loans, corporations also receive debt financing from the corporate bond market. For some firms, it is either impossible or costly to issue corporate bonds in international markets (see Massa and Žaldokas 2014). The preference of investors for local bonds could result from differences in taxation (e.g., the absence of withholding taxes on offshore bonds). In addition,

[16] Some debt contract features included as control variables could be endogenous with foreign lender participation. Therefore, they might be correlated with the error terms. In unreported tests, I also estimate a probit model without loan-level control variables as an alternative specification, and I find that the results are quantitatively similar.

Table 3
International corporate diversification and international debt financing

A. Baseline regression

	(1)	(2)	(3)	(4)	(5)
	Probit		OLS		Probit
Dependent variable:	Have foreign lender	Have foreign lead lender	Foreign lender share	Foreign lead lender share	International bond
GlobalDiv	0.133***	0.041***	0.040***	0.014***	0.052**
	(0.021)	(0.013)	(0.007)	(0.005)	(0.026)
Log(sales)	0.030***	0.012**	0.019***	0.006**	0.012
	(0.010)	(0.006)	(0.004)	(0.003)	(0.011)
Sales growth	0.004	0.001	0.001	0.000	0.031
	(0.002)	(0.001)	(0.001)	(0.001)	(0.019)
Cash flows	−0.192	−0.203	−0.160**	−0.055	−0.270
	(0.308)	(0.157)	(0.079)	(0.059)	(0.366)
Market-to-book	0.009	0.003	0.002	0.000	−0.013
	(0.008)	(0.005)	(0.002)	(0.002)	(0.014)
Leverage	0.179***	0.084***	0.058***	0.037***	0.031
	(0.044)	(0.024)	(0.015)	(0.011)	(0.066)
Cash	0.171*	0.107**	0.052*	0.037	−0.006
	(0.102)	(0.051)	(0.030)	(0.024)	(0.140)
STD(cash flows)	0.082	0.363	0.236	0.136	−0.438
	(0.583)	(0.325)	(0.155)	(0.113)	(0.662)
Log(facility/proceed amount)	0.046***	0.003	0.008**	0.006**	0.141***
	(0.008)	(0.005)	(0.003)	(0.003)	(0.012)
Log(maturity)	0.035**	0.039***	0.012***	0.011***	0.015
	(0.015)	(0.008)	(0.005)	(0.004)	(0.017)
Secured	−0.001	0.003	0.010	−0.014**	−0.049
	(0.026)	(0.014)	(0.009)	(0.006)	(0.038)
Missing secured	0.093***	0.026*	0.055***	0.007	
	(0.022)	(0.013)	(0.008)	(0.005)	
Log(number of lenders)	0.404***	0.020***	0.034***	−0.033***	
	(0.015)	(0.006)	(0.005)	(0.004)	
Private debt					−0.299***
					(0.021)
Callable					0.114***
					(0.028)
Rating, industry, quarter FE	Yes	Yes	Yes	Yes	Yes
Loan purpose, loan type FE	Yes	Yes	Yes	Yes	No
Observations	11,639	11,639	11,639	11,639	3,406
Pseudo/Adj R-squared	0.460	0.122	0.272	0.0938	0.476

(*continued*)

the location of global investors in corporate bond markets could influence the information to which they have access. For this reason, even if the bonds are unsecured or if foreign assets are not used as collateral, having foreign subsidiaries could potentially improve a firm's ability to issue corporate bonds in international markets.

Using the sample of corporate bonds from SDC, I test the hypothesis that multinationals have broader access to global investors in corporate bond markets than domestic firms. I estimate a probit model in which the dependent variable is an indicator variable equal to one if a bond is an international bond and zero otherwise. This specification models a firm's decision to place the bond in international markets as a function of *GlobalDiv* and a set of control variables. The same firm-specific control variables used in the bank

Table 3
Continued

B. Propensity-score-matching estimation

	Average treatment effect			
	Multinational (treated)	Domestic (control)	Difference (SE)	
Have foreign lender	0.639	0.535	0.104	(0.015)***
Have foreign lead lender	0.163	0.110	0.053	(0.010)***
Foreign lender shares	0.212	0.142	0.070	(0.006)***
Foreign lead lender shares	0.041	0.027	0.014	(0.004)***
Paired bank loan observations	5,790			
International bond	0.356	0.309	0.047	(0.016)***
Paired corporate bond observations	1,473			

This table presents the estimates of the effect of international corporate diversification on foreign lender participation using the sample of bank loan issuance from DealScan and international bond issuances using the sample of corporate bond issuance from SDC. *GlobalDiv* is defined as an indicator variable, and it takes the value of one if a firm's foreign pretax income is not missing in at least one year over the previous three years and there is at least one subsidiary in a country outside the United States. Panel A presents the probit and OLS estimates, and it uses the full sample of bank loan issuances at the facility level and corporate bond issuances. Columns 1 and 2 report the marginal effects at means of variables from probit regressions in which the dependent variables are an indicator denoting at least one foreign lender (Column 1) and whether a lead bank is a foreign lender (Column 2). Columns 3 and 4 present OLS regressions in which the dependent variables are the share of foreign lenders (Column 3) and the share of foreign lead lenders (Column 4). In Column 5, the dependent variable is the international bond issuance, and marginal effects from a probit regression are reported. In all columns, firm characteristics and loan (or bond) characteristics are included as controls. I also implement loan type and loan purpose fixed effects, S&P credit rating fixed effects, industry fixed effects at the one-digit SIC industry level, and quarter fixed effects. Panel B presents propensity-score score-matching estimation. The table reports the average difference of foreign lenders' participation between bank loans to multinationals and those to matched domestic firms, and it reports the average difference of international bond issuances between bonds issued by multinationals and those to matched domestic firms. Propensity scores are obtained from the probit model predicting whether the firm is multinational. Control variables in the probit model include firm-specific and loan-specific (bond) characteristics, like in Columns 1 and 5 of panel A. Using propensity score estimation with replacement, each loan (bond) of a multinational (treated group) is matched to the loan (bond) of a domestic firm (control group) within the same S&P credit rating category in the same quarter. I apply the nearest neighbor matching estimator, like in Becker and Ichino (2002) of 0.1 caliper, imposing the common support condition and bootstrapped errors with 500 replications. Table A1 (see the appendix) defines the variables. The standard errors with clustering at the firm level are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

loan regressions are included, and bond-specific variables, such as *Log(proceed amount), Log(maturity), Secured*, *Private debt,* and *Callable*, are additionally controlled.

The marginal effects from the probit regressions are reported in Column 5. The coefficient of *GlobalDiv* is positive and statistically significantly different from zero, which implies that the likelihood of an international bond issuance is positively associated with the issuer being multinational. Being multinational increases the probability of issuing a bond in international markets by 5.2 percentage points. The increase is equivalent to a 23% increase in the predicted probability of issuing a bond in international markets by a domestic firm.

2.2 Propensity-score-matching estimation

Given the differences between multinationals and domestic firms, I address the potential concern that multinationals' better access to foreign lending markets occurs because of differences in characteristics of the two groups. To address this self-selection issue, I construct a matched sample based on observable factors affecting the likelihood of being a multinational. To do so, I implement the propensity score matching proposed by Dehejia and Wahba (2002). First, using the bank loan sample, I calculate each loan's propensity score from a probit regression that estimates the likelihood that a firm is multinational. When estimating this equation, I include all firm-specific and loan-specific controls used in panel A of Table 3. After estimating the propensity score for each loan of a multinational (treated group), I find one matching loan of a domestic firm (control group) with the closest propensity score within the same S&P credit rating categories in the same quarter with replacement.[17] I then measure the difference between the foreign lender participation of multinationals and that of matched domestic firms, which is computed as the average treatment effects.

I report the results from these matching estimators using the 5,790 paired observations for bank loans in panel B of Table 3. The matching procedure finds a comparable domestic firm for each multinational firm.[18] Matching this way, the difference in foreign lender participation variables is significant for the loans to multinationals compared to their propensity-score-matched domestic firms. On average, 64% of multinationals have bank loans from foreign lenders, and 16% of multinationals have foreign lenders as a lead bank. In contrast, 54% and 11% of domestic firms have bank loans from foreign lenders and foreign lead lenders, respectively. The foreign lender share and foreign lead lender share of multinationals are 7% and 1% higher, respectively, than those of domestic firms. The differences between multinational and domestic firms are statistically significant.[19]

The matched sample for corporate bonds is constructed in a similar way. For each bond issued by a multinational, I find a matching bond issued by a domestic firm that is closest to the multinational firm in the same quarter within the same S&P credit rating category. This process leads to a matched corporate bond sample consisting of 1,473 paired observations. I find that conditional on issuing corporate bonds, the probability of issuing international bonds is 4.7 percentage points higher for multinationals than for comparable domestic firms. The magnitude of the effect of the multinational status on foreign lender participation and international bond issuance from the propensity score estimation is similar to that from the baseline regressions in panel A of Table 3.

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[17] Since the number of loans in a treated group (loans to multinational) is larger than that of control group (loans to domestic firms), matching with replacement can draw the same observations repeatedly from the control group as a match. Although matching without replacement can improve the precision of estimates, imposing the restriction of no replacement dramatically reduces the number of paired observations in a matched sample and the order of observations can affect the sample of loans or bonds that are finally matched. See Dehejia and Wahba (2002) for a detailed discussion of the costs and benefits of matching with and without replacement.

[18] Table A2 (see the appendix) reports the probit models estimating the propensity scores with pre-match and post-match samples. The coefficients on the firm-specific variables and loan-specific (bond-specific) variables are not statistically significant in the probit models predicting an issuance by multinational firms when I use the post-match sample.

[19] These findings using the propensity-score score-matching estimation are robust if sales is replaced with total assets.

2.3 Location of foreign subsidiaries and the intensity of foreign operations

The evidence documented so far supports the view that multinationals use more funding from lenders from outside the United States than domestic firms. However, the financial advantage of international corporate diversification could potentially occur through an alternative channel. If cash flows from different operating markets are not perfectly correlated, multinational firms tend to have lower cash flow volatility than domestic firms, which may lead to superior overall debt capacity.[20] However, if multinationals have better access to both domestic and foreign lenders because of their low credit risk, one would not expect that they have greater access to foreign lenders from the countries in which foreign subsidiaries are located. If enhanced access to foreign capital markets is mainly driven by the location of multinationals' foreign businesses, rather than their lower cash flow volatility, variation in the location of foreign subsidiaries should explain the availability and its sources of foreign funding within multinationals. In this section, I examine the impact of the location of foreign subsidiaries and the intensity of foreign operations on multinationals' sources of capital in three ways.

First, using a subsample of bank loans only to multinationals, I identify the locations of both foreign subsidiaries and lenders by region. Then, I consider whether the existence of foreign operations in each region increases the probability of a multinational having lenders from the same region. I focus on three regions (North America, Europe, and Asia) because foreign lenders from those countries are the most active investors in bank loan markets to U.S. firms. Since more than 60% of multinational firms have a subsidiary in the United Kingdom, I exclude the United Kingdom in this analysis when identifying European subsidiaries and lenders. I estimate a probit model in which the dependent variable is an indicator variable equal to one if the loan has at least one lender from each region. Instead of *GlobalDiv*, the regressions include dummy variables indicating whether the multinational has a subsidiary in Canada, Europe, Asia, Latin America, or the Middle East, respectively.

The results in panel A of Table 4 show that having operations in Europe increases the probability of having lenders from the same region by 12.1 percentage points, but it has little impact on the probability of having lenders from Asia. Similarly, having operations in Asia leads to a 6.8-percentage-point higher probability of having lenders from Asian countries, but it has negligible effect on borrowing from European lenders.[21] I also perform a joint test whether the coefficients on (1) *Sub in Canada* in Column 1, (2) *Sub in*

[20] The same logic in the large literature on industrial diversifications can be applied to international diversifications. It is called the "debt coinsurance effect," which was first noted by Lewellen (1971). For empirical evidence, see Berger and Ofek (1995).

[21] The coefficients of *Sub in Canada* in Column 2 and *Sub in Latin America* in Column 3 are significantly negative. Given that the syndicate loan amount is contributed by the limited number of lenders, these results are consistent with the substitution effects across lenders from different regions. For example, having a subsidiary in Europe

Table 4
International corporate diversification and international debt financing: Location of foreign subsidiaries and intensity of foreign operations

A. Location of subsidiaries by region

	(1)	(2)	(3)	(4)	(5)	(6)
	Probit			OLS		
	Have lender from:			Shares of lenders from:		
Dependent variable:	Canada	Europe	Asia	Canada	Europe	Asia
(A) Sub in Canada	0.042*	−0.056*	0.021	−0.000	−0.011	0.001
	(0.024)	(0.031)	(0.025)	(0.003)	(0.007)	(0.004)
(B) Sub in Europe	−0.028	0.121***	0.037	−0.005*	0.022***	−0.001
	(0.027)	(0.036)	(0.029)	(0.003)	(0.008)	(0.005)
(C) Sub in Asia	−0.042	0.048	0.068**	−0.002	0.011	0.008*
	(0.027)	(0.034)	(0.028)	(0.002)	(0.007)	(0.005)
Sub in Latin America	0.000	0.025	−0.043*	−0.001	0.006	−0.007*
	(0.025)	(0.030)	(0.025)	(0.002)	(0.007)	(0.004)
Sub in Middle East	0.021	0.025	0.026	0.001	0.002	0.006
	(0.026)	(0.034)	(0.028)	(0.002)	(0.008)	(0.005)
Observations	6,455	6,455	6,455	6,455	6,455	6,455
Pseudo/Adj R-squared	0.296	0.414	0.429	0.0167	0.189	0.150
Joint test	(A) Sub in Canada = 0 and (B) Sub in Europe = 0 and (C) Sub in Asia = 0			(A) Sub in Canada = 0 and (B) Sub in Europe = 0 and (C) Sub in Asia = 0		
Chi-squared (p-value)		22.08 (.0001)			31.97 (.0000)	

B. Intensity of foreign operations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS						
Dependent variable:	Foreign lender shares						
GlobalDiv	0.042***	0.011	0.002	0.021**	0.018*	0.017*	0.026***
	(0.008)	(0.009)	(0.011)	(0.010)	(0.010)	(0.009)	(0.010)
GlobalDiv x % foreign income	0.022***						
	(0.005)						
GlobalDiv x % foreign sales		0.127***					
		(0.025)					
GlobalDiv x HHI(sales)			0.094***				
			(0.018)				
GlobalDiv x Sub high private credit				0.029***			
				(0.011)			
GlobalDiv x Sub strong creditor right					0.034***		
					(0.010)		
GlobalDiv x Sub distant						0.039***	
						(0.010)	
GlobalDiv x Sub unfamiliar							0.024**
							(0.010)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,213	11,105	11,102	11,639	11,639	11,639	11,639
Adj R-squared	0.129	0.127	0.126	0.273	0.274	0.275	0.273

The table presents the effect of multinationals' foreign subsidiary locations and the intensity of foreign operations in international debt financing. Panel A presents results from regressions estimating the effect of location of subsidiaries by region on the nationality of foreign lenders in bank loans. The sample is restricted to the loans to multinationals. I report marginal effects from probit models estimating the probability of having lenders from Canada in Column 1, from Europe in Column 2, and from Asia in Column 3. In Columns 4 and 6, I report estimates of OLS regressions in which the dependent variables are lender shares from those regions. The main explanatory variables are indicators of having a subsidiary in each region (Europe, Asia, Canada, Latin America, and the Middle East). Since more than 60% of multinationals have a subsidiary in the United Kingdom, having a subsidiary in the United Kingdom is not considered in European lender participation. Panel B presents estimates from OLS regressions estimating the effect of the intensity of foreign operations on foreign lender shares. Regressions additionally include the interaction term between *GlobalDiv* and the measures for the intensity of foreign operations and the foreign subsidiary country characteristics. Control variables include firm, loan, and bond characteristics like in Columns 2 and 5 in Table 3. The coefficients of control variables are not reported to save space. Table A1 (see the appendix) defines the variables. The standard errors with clustering at the firm level are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Europe in Column 2, and (3) *Sub in Asia* in Column 3 are jointly significant. The chi-square statistic is significant at the 1% level, which implies that having a subsidiary in a specific region increases the probability of having a foreign lender from the same region. I also test the effect of multinationals foreign subsidiary locations at the country level. In Table A3 (see the appendix), I find that having subsidiaries, especially in Canada, Japan, Germany, Netherlands, Belgium, or Australia, has a significantly positive impact on the participation of lenders from the same country. One thing to note is that the results are not driven by the lenders from Switzerland and Hong Kong, both of which have globally open financial markets.

Second, I examine whether the intensity of foreign operations strengthens multinationals' access to international capital markets. I measure the degree of global diversification in three ways: the proportion of foreign income to total income, the proportion of foreign sales to total sales, and one minus the concentration of foreign sales. I report baseline OLS regressions in which the dependent variable is foreign lender shares, like in panel A of Table 3. The regressions additionally include *GlobalDiv* interacted with the measures of the degree of global diversification. The results are reported in Columns 1 to 3 of panel B of Table 4. The coefficients of the interaction terms are positive and statistically significant. These results suggest that when multinationals are more internationally diversified, they are more likely to rely on foreign lender loans.

Third, I exploit the characteristics of the countries of foreign subsidiaries. If the existence of foreign operations is the reason for improved access to foreign debt financing, the difference in access to foreign bank loans between multinationals and domestic firms should vary with financial market conditions and the cost of borrowing in countries in which multinationals' foreign subsidiaries are located. In Columns 4 and 5 of panel B of Table 4, I find that the existence of subsidiaries in countries with more private credit available and with stronger protection on creditor rights significantly increases foreign lender shares. The results in Columns 6 and 7 indicate that the difference in foreign lender shares between multinational and domestic firms substantially increases if multinationals operate in countries that are physically distant from the United States and in unfamiliar countries (as measured by trade openness). These findings confirm the financial channel through which the physical operational presence in financially developed countries improves access to foreign bank lending. In addition, the results are consistent with the view that the operation presence can considerably reduce any financial frictions in the countries in which monitoring and information acquisition might be costly.

increases the likelihood of borrowing from European lenders, while it consequently decreases the probability of including lenders from other regions.

3. Financial Flexibility during the 2007–2009 Financial Crisis

3.1 Multinational effect on financial policy during the financial crisis

In the previous section, I document that by having operations in other countries, multinational firms can diversify their sources of funding across countries. Diversification in capital sources can help multinationals hedge against disruptions in a particular capital market. If there is a credit supply shock in their home country, multinationals are more able than domestic firms to shift to alternative funding channels in other countries that are less affected. Consequently, when one of the financing channels in a specific country is impaired, the investment decisions of internationally diversified firms are less adversely affected than those of domestic firms.

The recent 2007–2009 financial crisis provides a useful setting to test these predictions. A number of papers have documented that during the financial crisis, banks sharply cut their lending and increased loan interest rates to corporate sectors.[22] Therefore, firms that could not receive enough funding to finance their investments were forced to utilize alternative capital markets. Otherwise, they had to forgo valuable NPV projects and cut their investment.[23] If lenders in other countries were less affected by the supply shock, multinationals could potentially cope better by raising capital abroad. On the other hand, domestic firms that had extensively relied on domestic capital markets might experience difficulties in finding alternative funding sources outside the United States. Accordingly, the difference in foreign debt issues between multinationals and domestic firms would increase, especially during the financial crisis.

One feature of the 2007–2009 financial crisis is that it was not localized to one market but was a global shock that also affected non-U.S. financial markets. However, not all countries were similarly affected by the financial crisis. Recent papers have documented a substantial cross-lender and cross-country variation of banks' performance during the financial crisis based on their financing structure or county-level regulations (see, e.g., Beltratti and Stulz 2012; Erkens, Hung, and Matos 2012). To the extent that lenders in different countries were affected differentially, the ability of

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22 For example, Ivashina and Scharfstein (2010) use data from the syndicated loan market to argue for evidence of a contraction in bank credit availability during the peak of the crisis. In addition, Santos (2011) uses data of bank loans from DealScan to show that borrowers took smaller loans and paid higher loan spreads during the crisis period, which indirectly supports the claims on reduced credit availability. However, the evidence on the causal effect of the recent crisis on the reduction in bank lending to firms is not yet conclusive. Using a different set of data, Chari, Christiano, and Kehoe (2008) and Boyson, Helwege, and Jindra (2014) conclude that bank lending was not reduced as expected during the crisis.

23 Several empirical studies provide evidence of the real effect of the 2007–2008 financial crisis on corporate sectors. Campello, Graham, and Harvey (2010) conduct a survey of 574 CFOs of U.S. firms. The CFOs who think their firms are financially constrained state that the firms had difficulties in extending their credit, and as a result, cut their investment during the financial crisis. Duchin, Ozbas, and Sensoy (2010) document that firms without enough cash reserves before the crisis experienced a sharp reduction in capital expenditures. However, not all empirical evidence reaches the same conclusion regarding the causal effect of the credit contraction from 2007 to 2008 on firm's financing and investment policies (see Kahle and Stulz 2013).

multinationals to obtain financing from multiple sources was still potentially beneficial.

Using the 2007–2009 financial crisis as an exogenous shock to the supply of capital, I construct a test of whether easier access to foreign funding sources allows multinationals to be more financially flexible. In particular, I compare capital raising activities of multinationals and domestic firms to examine whether multinationals were more likely to raise funding in foreign capital markets during the financial crisis more than before relative to domestic firms. To test this hypothesis, I use a specification similar to the one used in panel A of Table 3, but include crisis indicator variables and interaction terms between the crisis indicator variables and *GlobalDiv*. Because I want to capture the effect of the crisis period, I omit quarter fixed effects but do control for firm and loan characteristics used in panel A of Table 3.

If multinationals rely more on funds from global capital sources than domestic firms, especially during the crisis, then the coefficient of the interaction terms should be significantly positive. After the bankruptcy of Lehman, funds to corporate sectors became less available, and the lending costs to the corporate sector increased sharply, which negatively affected financing and operations of U.S. firms. If foreign lenders were less affected by the financial crisis and only the firms with access to foreign lenders were able to take advantage of receiving necessary funding, the difference in foreign lender participation between multinationals and domestic firms would increase during the financial crisis.

The results for bank loans are reported in panel A of Table 5. If I look at foreign lenders only in Columns 1 and 3, the positive effect of being multinational on foreign lender participation is not significantly stronger in the crisis period. However, in the regressions in Columns 2 and 4, the coefficients of the interaction term between *GlobalDiv* and *Late crisis* are positive and significant for foreign lead lender participations. During the noncrisis period, loans to multinationals were 3.7 percentage points more likely to have a foreign lead lender than domestic firms. This difference in foreign lead lender participation between multinationals and domestic firms sharply increased during the two quarters after Lehman's collapse. While domestic firms experienced a 5.2 percentage points decrease in the probability of having a foreign lead lender, multinationals firms were 23.1 percentage points more likely to receive loans from a foreign lead lender than domestic firms during that period.[24] In the OLS regression in Column 4, I find that the foreign lead lender share for

[24] The interpretation on the interaction terms in probit models is econometrically challenging. Norton, Wang, and Ai (2004) emphasize that it is difficult to interpret the interaction effects in nonlinear models, such as the probit model used here, because the interaction effects can be different for each observation point of independent variables. I report the average interaction effects of the estimates on the interaction terms in the bottom of the table. For the regression in Column 2, the mean interaction effect of *GlobalDiv*Late crisis* has the average z-statistics of 1.995. When I look at the mean interaction effect, the magnitude the interaction effect decreases, but the interaction effect is mostly positive and significant in the range of higher predicted probabilities.

Table 5
International corporate diversification and financial policy during the financial crisis

A. Baseline regression

	(1)	(2)	(3)	(4)	(5)
	Bank loans				Corporate bonds
	Probit		OLS		Probit
Dependent variable:	Have foreign lender	Have foreign lead lender	Foreign lender shares	Foreign lead lender shares	International bond
GlobalDiv	0.129***	0.037***	0.040***	0.011**	0.044
	(0.022)	(0.013)	(0.007)	(0.005)	(0.028)
Early crisis	−0.079	0.019	0.009	0.006	0.484***
	(0.093)	(0.054)	(0.025)	(0.017)	(0.134)
Early crisis x GlobalDiv	0.024	0.019	−0.001	0.021*	0.018
	(0.058)	(0.035)	(0.015)	(0.011)	(0.058)
Late crisis	−0.177	−0.052	0.015	0.020	0.431***
	(0.137)	(0.043)	(0.028)	(0.024)	(0.138)
Late crisis x GlobalDiv	0.138	0.231**	0.013	0.079**	0.237*
	(0.112)	(0.114)	(0.033)	(0.031)	(0.124)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	11,639	11,639	11,639	11,639	3,406
Pseudo/Adj R-squared	0.460	0.123	0.271	0.0954	0.477
Mean interaction effect for					
Early crisis x GlobalDiv	0.015	0.008			0.015
	[0.129]	[0.207]			[0.115]
Late crisis x GlobalDiv	0.081	0.171			0.150
	[0.888]	[1.995]			[1.263]

(*continued*)

Table 5
Continued

B. Propensity-score-matching estimation

	Have foreign lender					Foreign lender share			
	Paired obs.	Multinational	Domestic	Difference (SE)		Multinational	Domestic	Difference (SE)	
(1) Pre-crisis	4,633	0.6518	0.5444	0.1075	(0.017)***	0.2180	0.1434	0.0746	(0.007)***
(2) Early crisis	578	0.5779	0.5138	0.0640	(0.049)	0.1787	0.1227	0.0560	(0.017)***
(3) Late crisis	115	0.4609	0.1739	0.2870	(0.097)***	0.1449	0.0614	0.0835	(0.038)**
(2) Early crisis - (1) Pre-crisis				–0.042	(0.027)			-0.018	(0.012)
(3) Late crisis - (1) Pre-crisis				0.179	(0.056)**			0.009	(0.027)

	Have foreign lead lender					Foreign lead lender share			
	Paired obs.	Multinational	Domestic	Difference (SE)		Multinational	Domestic	Difference (SE)	
(1) Pre-crisis	4,633	0.1489	0.1012	0.0477	(0.011)***	0.0343	0.0253	0.0090	(0.004)**
(2) Early crisis	578	0.1747	0.0761	0.0986	(0.025)***	0.0540	0.0212	0.0328	(0.013)**
(3) Late crisis	115	0.2870	0.1130	0.1739	(0.088)*	0.0781	0.0174	0.0607	(0.022)***
(2) Early crisis - (1) Pre-crisis				0.051	(0.021)*			0.024	(0.009)**
(3) Late crisis - (1) Pre-crisis				0.126	(0.049)*			0.052	(0.021)*

	International bond				
	Paired obs.	Multinational	Domestic	Difference (SE)	
(1) Pre-crisis	1,025	0.1707	0.1366	0.0341	(0.015)**
(2) Early crisis	119	0.6807	0.5042	0.1765	(0.066)***
(3) Late crisis	56	0.8750	0.8393	0.0357	(0.069)
(2) Early crisis - (1) Pre-crisis				0.142	(0.062)*
(3) Late crisis - (1) Pre-crisis				0.002	(0.073)

This table presents the changes in the effect of international corporate diversification on access to foreign lenders in bank loans and international corporate bond markets over the 2007–2009 financial crisis periods. Panel A reports estimates of probit and OLS regressions in which international debt financing variables are regressed on an indicator variable of multinational firms as well as interactions between the multinational indicator variable and two crisis period indicator variables. *Early crisis* is defined as calendar quarters between 2007Q3 and 2008Q3 and *Late crisis* as 2008Q4-2009Q1. In Columns 1 to 4, the estimates of probit and OLS regressions using the bank loan sample are reported, and the regressions include firm and loan characteristics as control variables as is done in Column 1 of Table 3. Column 5 reports estimates of probit regressions in which a dependent variable is an indicator variable denoting an international bond. Control variables include firm and bond characteristics as is done in Column 5 of Table 3. On the bottom of the table, I report the mean interaction effects and z-statistics corrected by the methodology of Norton, Wang, and Ai (2004). Panel B reports the average differences in foreign lender participation and international bond issuance between the multinationals and the matched domestic firms by subperiod: *Pre-crisis* (2000Q1-2007Q2), *Early crisis* before the collapse of Lehman (2007Q3-2008Q3), and *Late crisis* after the collapse of Lehman (2008Q4-2009Q1) using the propensity-score score-matching sample. Using a propensity-score score-matching procedure, each loan (bond) of a multinational (treated group) is matched to the loan (bond) of a domestic firm (control group). The coefficients of control variables are not reported to save space. Table A1 (see the appendix) defines the variables. The standard errors with clustering at the firm level are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

domestic firms did not change in the last two quarters of crisis, compared to the noncrisis period. However, the foreign lead lender share for multinationals was 7.9 percentage points higher in the later crisis period than in the noncrisis period.

In Column 5, I estimate similar regressions using the sample of corporate bonds. The coefficient on *Early crisis* is significant and positive, but the coefficient of the interaction term, *Early crisis* x *GlobalDiv*, is insignificant, suggesting that both multinationals and domestic firms actively used international bond markets in the early crisis period. Given that firms that are able to issue corporate bonds are relatively large firms, the result using the corporate bond issuance sample implies that corporate bond markets are relatively more integrated with less friction. However, I find evidence that the difference between multinational and domestic firms in the probability of international bond issuance sharply increased in the two quarters after September 2008.

In panel B of Table 5, I estimate changes in the effect of being multinational during the crisis period using a propensity score matched sample. I report the average difference of foreign lender participations and international bond issuance between multinationals and comparable domestic firms separately for subperiods. The difference in foreign lender participation between multinationals and domestic firms became 18 percentage points larger in the crisis period after Lehman's collapse, which is nearly three times the difference in the pre-crisis period. The difference in foreign lead lender participation across the two groups is five times larger in the two quarters after the fourth quarter of 2008 compared to the average difference of 4.8% during the pre-crisis period. Using the propensity score matched sample of bank loan and bond issuances, this comparison confirms the main finding that multinationals were more able to raise debt capital abroad than domestic firms, particularly when the supply of capital in the United States was impaired.

3.2 Sources of bank loans to multinational firms during the crisis

I further examine whether the location of foreign operations can explain the source of the increase in bank loan issuances by multinationals from foreign lenders during the financial crisis. I take a sample of 6,455 loans issued only by multinationals and examine how multinationals' access to lenders, particularly from countries in which they have foreign subsidiaries, changes over the crisis period. In Columns 1 to 4 in Table 6, I find that the probability of borrowing from lead lenders from countries in which multinationals have foreign subsidiaries increased by 27 percentage points in the two quarters after Lehman's collapse. In addition, the proportion of loan amounts retained by foreign lenders from the same countries in which multinationals have foreign subsidiaries significantly increased by 3 percentage points in the crisis period after Lehman.

Table 6
Foreign lender participation of loans issued by multinationals during the financial crisis: Location of foreign subsidiaries

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Probit		OLS		Probit		OLS	
	From countries in which multinationals have foreign subsidiaries:							
Dependent variable:	Have foreign lender	Have foreign lead lender	Foreign lender shares	Foreign lead lender shares	Have foreign lender	Have foreign lead lender	Foreign lender shares	Foreign lead lender shares
Early crisis	−0.026	0.077	0.025	0.022	−0.006	0.045	0.027	0.041
	(0.102)	(0.066)	(0.025)	(0.015)	(0.127)	(0.105)	(0.044)	(0.035)
Late crisis	−0.002	0.276**	−0.004	0.030*	−0.346**	−0.046	−0.085*	−0.045
	(0.108)	(0.116)	(0.024)	(0.016)	(0.174)	(0.097)	(0.045)	(0.037)
Less affected sub					0.069**	0.019	0.015	0.006
					(0.035)	(0.024)	(0.012)	(0.008)
Early crisis x Less affected sub					−0.160	−0.029	−0.050	−0.036
					(0.108)	(0.056)	(0.034)	(0.030)
Late crisis x Less affected sub					0.206***	0.309	0.094**	0.093**
					(0.048)	(0.200)	(0.046)	(0.042)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,455	6,455	6,455	6,455	6,455	6,455	6,455	6,455
Pseudo/Adj R-squared	0.389	0.126	0.239	0.0805	0.462	0.121	0.265	0.123
Mean interaction effect for								
Early crisis x Less affected sub					−0.117	−0.098		
					[−1.591]	[−1.119]		
Late crisis x Less affected sub					0.301	0.400		
					[2.039]	[2.711]		

This table presents the effect of the 2007–2009 financial crisis on foreign lender participation of loans issued by multinational firms depending on their foreign subsidiary locations. The sample is restricted to the bank loans issued by multinationals. The table reports estimates of probit and OLS regressions in which the dependent variables are the indicator variables of receiving a bank loan from at least one foreign lender from countries in which a multinational firm has subsidiaries in Columns 1 and 2 and the shares retained by lenders from countries in which the firm has subsidiaries in Columns 3 and 4. Columns 5 to 8 report the estimates of probit and OLS regressions, where the dependent variables are foreign lender participation variables. All regressions include firm and loan characteristics as control variables, as is done in Column 1 of Table 3, but the coefficients of control variables are not reported to save space. *Early crisis* is defined as calendar quarters between 2007Q3 and 2008Q3, and *Late crisis* as 2008Q4-2009Q1. *Less affected sub* is defined as one if a multinational firm has at least one subsidiary in the countries in which the change in the ratio of private credit to GDP between 2007 and 2009 is above the top tercile among 53 countries. On the bottom of the table, the mean interaction effects and z-statistics corrected by the methodology of Norton, Wang, and Ai (2004) are reported. Table A1 (see the appendix) defines the variables. The standard errors with clustering at the firm level are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

In addition, I expect that if multinationals shifted to alternative funding sources during the financial crisis because of tight U.S. credit markets, then these firms should find financing from the countries in which its financial market is less impaired during the 2007–2009 financial crisis. I evaluate this hypothesis in Columns 5 to 8 in Table 6. The effect of the financial crisis at the country level is measured by the changes in the private credit to GDP ratio over 2007 to 2009, and *Less affected sub* is an indicator variable equal to one if a multinational company has a subsidiary in a country that experienced the top 33% positive (or nonnegative) changes in its private bank lending.[25] In addition to the crisis indicators, I include *Less affected sub* indicator and the interaction term of this variable with the crisis dummy variables. The interaction term, *Late crisis x Less affected sub*, is statistically significant in three of the four columns, Column 6 notwithstanding. These results suggest that multinational firms were able to receive bank loans only if they had subsidiaries in countries in which bank lending was not severely diminished during the crisis.

To see which types of firms drive this result, in Table A4 (see the appendix), I separate the loans based on the borrower's S&P credit ratings. I find a stronger result that multinationals used loans from foreign lenders during the financial crisis especially for the borrowers without a public bond rating. Those firms are potentially affected more by the recent financial crisis, so they have to seek funding from alternative channels, which includes looking outside the United States. This result is consistent with the prediction that diversification in capital sources is particularly beneficial for financially constrained and bank-dependent firms.

4. Investment Policy during the 2007–2009 Financial Crisis

4.1 Multinational effect on investment during the financial crisis

Access to foreign funding only matters if a lack of access to funding creates real consequences for firms' investment activities. I examine the difference in investment activities between multinationals and domestic firms before and after the financial crisis. If multinationals' access of foreign funding sources alleviates the adverse effect of the credit supply shock in the United States, their investment policies would be less affected than if they used only domestic funding sources.

[25] This measure of the extent to which each country was adversely affected by the 2007–2009 financial crisis is motivated by Claessens et al. (2010). In the sample of 53 countries, on average, private credit/GDP ratios changed by 2.7% from 2007 to 2009, and the United States experienced a sharp decrease of 7.1%, which is in the bottom 10% of the distribution. The countries that are less affected by the financial crisis include Brazil, Canada, Chile, Czech Republic, Hungary, Ireland, New Zealand, Peru, Poland, Portugal, Romania, Singapore, Slovenia, South Korea, Spain, and Turkey. As an alternative measure for the extent to which financial markets were affected at the country level during the financial crisis, I also use changes in stock market returns of financial firms between 2007 and 2009 at the country level. This alternative definition for *Less affected sub* does not change the statistical significance of the results.

One potential concern in the investment analysis is that the change in overall capital expenditures could be affected by demand changes rather than through the financing channel. It could be the case that multinationals' revenues from foreign operations do not change during the crisis, but domestic firms experience a sharp reduction in revenues in the United States, which would lead to less demand for debt than multinationals during the crisis.[26] To evaluate whether availability of funding affects investment, it is important to control for differential demand shocks of the financial crisis in domestic and foreign operations. For this reason, I compare the multinationals' investment on the U.S. operations to that of domestic firms. If foreign operations indeed made multinational firms more financially flexible, I would expect that the U.S. investment of multinationals was less affected than that of domestic firms during the financial crisis.

To compare the investment policy of multinationals and domestic firms during the financial crisis, I rely on two main investment variables: annual U.S. capital expenditures scaled by total assets and quarterly U.S. expansion announcements. Using the firm-year panel data, I estimate OLS regressions in which the dependent variable is the annual U.S. capital expenditures scaled by total assets.[27] As argued in Roberts and Whited (2012), some endogenous firm-level variables that are potentially correlated with measurement error in Tobin's q can lead to a biased result in investment regressions. For this reason, I include the following basic firm characteristics as control variables: *Log(sales), Sales growth, Cash flows*, and *Market-to-book*, which are all lagged by one year except *Cash flows*. Because the segment-level investment data are measured annually, I include the crisis period dummies indicating the calendar years of 2008, 2009, and 2010, respectively. All regressions include firm fixed effects, and standard errors are clustered at the firm level.[28]

The results are reported in panel A of Table 7. In Column 1, I find no evidence that U.S. capital expenditures of both multinationals and domestic

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[26] However, a decrease in demand could explain the reduction in lending overall market, but does not explain the previous result that firms increase lending abroad relative to lending at home markets.

[27] There could be a selection issue in the sense that multinational firms that report detailed segment-level capital expenditures information might be different from others that do not disclose. In untabulated results, I evaluate whether multinational firms that report segment-level capital expenditure variables are fundamentally different from those that do not report. I estimate the probability of reporting capital expenditure variables conditional on reporting geographical sales data in the Compustat Geographical Segment database. I find that smaller firms with higher cash flows and lower cash holdings are more likely to report segment-level capital expenditures, but the reporting behaviors do not vary over time (especially during the financial crisis period) after controlling for firm fixed effects. For this reason, it does not appear that selection bias is likely to be a serious concern in the investment regressions with firm fixed effects.

[28] Since in the firm-year panel regression I use a financial crisis as an experiment, which is a time-specific event, it is possible that the standard errors are correlated over time. As a robustness check, I also estimate the U.S. investment regressions with standard errors clustered jointly by firm and year. The statistical significance of the main results is robust with the two-way clustering. For consistency, the results with standard errors clustered at the firm level are reported.

Table 7
International corporate diversification and U.S. investment policy during the financial crisis

A. U.S. capital expenditure

	(1)	(2)
	OLS	
Dependent variable:	U.S. capital expenditures	
GlobalDiv	0.000	−0.002
	(0.005)	(0.005)
Crisis2008	−0.000	0.000
	(0.002)	(0.002)
GlobalDiv x Crisis2008	0.001	0.003
	(0.004)	(0.004)
Crisis2009	−0.021***	−0.021***
	(0.002)	(0.002)
GlobalDiv x Crisis2009	0.011**	0.011**
	(0.005)	(0.005)
Crisis2010	−0.018***	−0.017***
	(0.003)	(0.003)
GlobalDiv x Crisis2010	0.011*	0.011**
	(0.006)	(0.006)
(U.S.) Log(sales)	−0.008***	−0.011***
	(0.002)	(0.002)
(U.S.) Sales growth	0.081***	0.085***
	(0.009)	(0.008)
(U.S.) Cash flows	0.010***	0.013***
	(0.001)	(0.002)
(U.S.) Market-to-book	0.008***	0.012***
	(0.002)	(0.002)
Controls	Firm-level	U.S.-level
Firm FE	Yes	Yes
Observations	14,091	14,091
Adj R-squared	0.686	0.674

(*continued*)

firms significantly changed in 2008 relative to the pre-crisis period. In contrast, domestic firms reduced investment by 2.1 percentage points per year in 2009 and by 1.8 percentage points in 2010 relative to pre-crisis periods. On the other hand, multinationals reduced U.S. capital expenditures by only 1.0 and 0.7 percentage points during the same period. The differences are statistically significant.

Although I separately measure the multinational's U.S. investment as a dependent variable, one might still be concerned that looking at the U.S. investment does not perfectly control for the different demand shocks between multinational and domestic firms within the U.S. market. Because of this concern, I also estimate the regression with the control variables measured at the U.S. operation level. For each multinational firm, as control variables, I separately measure *Log(sales), Sales growth,* and *Cash flows* of multinationals' U.S. operations from the Compustat Geographic segment database and the median *Market-to-book* of domestic firms in the same industry (defined at the two-digit SIC code). In Column 2 of panel A, Table 7, I confirm the finding that multinational firms cut U.S. investment significantly less than domestic firms from 2009 to 2010, even after controlling for size of

Table 7
Continued

B. U.S. expansion announcements

	(1)	(2)	(3)	(4)
	OLS		Random effect probit	
Dependent variable:	U.S. expansion announcement			
GlobalDiv	−0.023**	−0.005	−0.014	0.040
	(0.009)	(0.016)	(0.022)	(0.036)
Early crisis	−0.012	0.006	−0.015	0.032
	(0.009)	(0.010)	(0.034)	(0.040)
GlobalDiv x Early crisis	0.049***	0.039**	0.115***	0.112*
	(0.010)	(0.016)	(0.035)	(0.058)
Late crisis	−0.071***	−0.045***	−0.268***	−0.198***
	(0.011)	(0.013)	(0.047)	(0.055)
GlobalDiv x Late crisis	0.082***	0.094***	0.288***	0.344***
	(0.013)	(0.023)	(0.051)	(0.082)
Post-crisis	−0.048***	−0.019*	−0.150***	−0.064
	(0.009)	(0.011)	(0.037)	(0.046)
GlobalDiv x Post-crisis	0.085***	0.057***	0.285***	0.184***
	(0.010)	(0.019)	(0.040)	(0.065)
(U.S.) Log(sales)	0.017***	0.004	0.192***	0.162***
	(0.004)	(0.007)	(0.006)	(0.009)
(U.S.) Sales growth	0.002	−0.001	0.043***	0.081**
	(0.004)	(0.008)	(0.017)	(0.038)
(U.S.) Cash flows	0.010	0.135***	0.244	0.683***
	(0.045)	(0.026)	(0.219)	(0.099)
(U.S.) Market-to-book	0.015***	0.031***	0.072***	0.104***
	(0.002)	(0.009)	(0.007)	(0.035)
Controls	Firm-level	U.S.-level	Firm-level	U.S.-level
Firm FE	Yes	Yes	Yes	Yes
Observations	107,333	49,296	107,333	49,296
Adj R-squared	0.189	0.180		
Log likelihood			−43,708	−1,9296
Wald chi-squared			2,859	970.3

This table reports results from panel regressions that estimate the changes in the effect of international corporate diversification on U.S. investment during the financial crisis period. In panel A, the sample includes firm-year panel observations from all domestic firms and multinational firms that reported nonmissing capital expenditure variables in the Compustat Segment database for the fiscal year ending from 2000 to 2010. Panel A reports estimates of OLS regressions with firm fixed effects in which the dependent variable is annual U.S. capital expenditure scaled by total assets. U.S. capital expenditure is calculated as capital expenditure of the U.S. segment for multinational firms from the Compustat Segment database. *Crisis2008, Crisis2009,* and *Crisis2010* are dummy variables indicating whether the fiscal year ended in 2008, 2009, and 2010, respectively. Control variables are estimated at the firm-level in Column 1 and at the U.S.-level in Column 2. The U.S.-level control variables are from the Compustat Segment database for multinational firms and from the Compustat Annual database for domestic firms. *GlobalDiv, Log(sales), Sales growth,* and *Market to-book*, except *Cash flows* are lagged by one period. In panel B, the sample includes firm-quarter panel observations from 2000Q1 to 2010Q4 of all publicly traded U.S. firms from the Compustat database. The dependent variables are the indicator variables of the quarters when the firm makes announcement of U.S. acquisitions or U.S. expansion. The estimates of OLS regressions with firm fixed effect are in Columns 1 and 2, and the estimates from random effect probit regressions are reported in Columns 3 and 4. *Early crisis* is defined as calendar quarters between 2007Q3 and 2008Q3, *Late crisis* as 2008Q4-2009Q1, and *Post-crisis* as 2009Q2-2010Q1. Table A1 (see the appendix) defines the variables. The standard errors clustered at the firm level are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

operations and growth opportunities in the U.S. markets during the crisis period.

In addition, to supplement the annual U.S. capital expenditure data, I consider U.S. expansion announcements based on corporate news, which are taken from

Capital IQ.[29] Using firm-quarter panel data from 2000 to 2010, I estimate an OLS model with firm fixed effects and a random effect probit model in which the dependent variable is the U.S. expansion announcement indicator. Since the dependent variable is a binary variable, including the interaction terms in the regression might complicate the interpretation, especially in panel data. For this reason, I focus on the OLS regression with firm fixed effects but present the estimates from a random effect probit model as a comparison. In the regressions, in addition to two crisis period indicators, I include *Post-crisis*, defined as the calendar quarters of 2009Q2-2010Q1, to capture the delayed effects on the investment policy.

The results are reported in panel B of Table 7. In Columns 1 and 3, with the firm-level control variables, I find that domestic firms experienced a sharp decrease in domestic expansions of 7.1 percentage points in the late crisis period and 4.8 percentage points post-crisis. On the other hand, the changes in multinationals' U.S. domestic expansion announcements were marginal during the late crisis and post-crisis periods. The difference in the response of the U.S. investment policy to the financial supply shock between multinationals and domestic firms is statistically significant. In Columns 2 and 4, although half of the observations are dropped because of the limited availability of segment-level cash flows items, the main results are robust with U.S. operation-level control variables. Given that the average quarterly domestic expansion announcement rate is 17%, the economic impact of being internationally diversified on the ability of firms to avoid capital supply shocks in their home country during the financial crisis appears to be substantial.

The investment regression results thus far provide consistent evidence of multinationals' financial flexibility during the financial crisis period, but a potential selection bias issue still remains. It is possible that internationally diversified firms reduced their investment much less than domestic firms during the financial crisis because they were the firms that were less financially constrained from the beginning regardless of their multinational status. I address this concern by performing propensity-score-matching estimations, which are similar to those for the foreign borrowing analysis in Section 3, for the investment analysis. I create the propensity-score-matching sample to see the average differences in U.S. capital expenditure and U.S. expansion announcements between multinational firms and the matched domestic firms

[29] The selection bias in the U.S. expansion announcement variable would be less severe, compared to the U.S. capital expenditure variable. Acquisition activities are obtained from the SDC database, which is known to provide fairly complete coverage of mergers and acquisitions transactions of the U.S. firms, especially in later years. If there is any, a selection bias on the U.S. expansion announcement variable is most likely to be caused by using the corporate expansion announcement information from Capital IQ, which is sourced from major news articles. In untabulated results, I find that the expansion announcement variable is a good proxy for the actual investment firms make, as it is highly correlated with the overall capital expenditure and acquisition accounting variables. In addition, with the firm fixed effects, the frequency of expansion announcements do not vary between multinationals and domestic firms. Thus, I think the selection bias would not be serious in the U.S. expansion announcement regressions with firm fixed effects.

during the financial crisis. Propensity scores are obtained from a probit model predicting whether the firm is multinational. Control variables in the probit model include *Log(sales), Sales growth, Cash flows,* and *Market-to-book*, which are estimated as of the end of fiscal year of 2006 (or 2006Q3). Using the propensity score estimation, each multinational firm is then matched to a domestic firm within the same S&P credit rating category in the same period (year or quarter). Matching within the rating category and with similar size partially alleviates the selection concern by controlling for the firms' financing ability that can be estimated by observable firm characteristics.

Table A5 (see the appendix) reports the results. When I compare the difference in the annual U.S. capital expenditure between multinationals and the matched domestic firms in panels A and B, the results are not as strong as those in the regression results in panel A of Table 7. The annual U.S. capital expenditure rate of multinationals was significantly lower than domestic firms by 3.1 percentage points in pre-crisis periods, but the difference decreased in 2009 and in 2010. In 2010, the difference between the multinationals and domestic firms becomes weakly significant or insignificant. In panels C and D, I compare the quarterly U.S. expansion announcement between multinationals and domestic firms by subperiods. Prior to the financial crisis, multinationals made slightly fewer U.S. expansion announcements, but in the late crisis and post-crisis periods (2008Q4-2010Q1), multinational firms were 4.5-5.5 percentage points more likely to expand in the United States than domestic firms, and the changes in differences are statistically significant.

The propensity-score-matching estimation in the investment regressions confirms that after controlling for the difference in overall financing ability between multinationals and domestic firms, the main results in the U.S. investment analysis still hold.

4.2 Cross-sectional differences in investment policy within multinationals

The differential impact of the financial crisis on U.S. investment between multinationals and domestic firms implies that international corporate diversification could help them access foreign capital markets when domestic markets are illiquid. Although I do control for growth opportunities and performance of firms' U.S. operations, it is still difficult to isolate that the main reason for the better performance of multinationals during the crisis is their superior ability to issue international debt. In this section, I examine whether the location of foreign subsidiaries of multinationals and the ability of issuing foreign debt can explain the different responses in domestic investment. The purpose of these tests is to understand whether the funding flexibility of multinationals is an important channel through which their domestic investment was less impaired than domestic firms during the crisis.

Like in panel B of Table 7, I construct a sample of firm-quarter observations of multinational firms and estimate a model predicting whether a firm makes an U.S. expansion announcement. In the regressions, I include *Less affected sub*, which is an indicator variable denoting that a multinational firm has a subsidiary in countries in which the private credits were less diminished during the crisis.[30] To estimate whether those multinationals indeed took advantage of foreign debt financing, I include *Issue foreign debt*, which is an indicator variable that takes a value of one if a multinational receives a bank loan from a foreign lead lender or if it issues international bond in a given quarter. The main interest is on the interaction terms of two variables with the crisis indicators.

Table 8 reports the estimates. The results presented in Columns 1 and 2 imply that multinational firms are less likely to make domestic expansion announcements, on average, by 3.6 and 8 percentage points in the early and late crisis periods, respectively. However, the U.S. expansion announcement rates are higher by 7.2 and 7.9 percentage points during the crisis periods, if the multinationals operate in countries in which bank credits were less affected by the financial crisis. These differences in the U.S. expansion announcement rates are statistically significant. When I consider multinationals' foreign debt issuance in Columns 3 and 4, the coefficients of the interaction terms, *Early crisis x Issue foreign debt* and *Late crisis x Issue foreign debt*, are statistically significant and positive. This result suggests that multinationals' investment policy was not affected by the financial supply shock as much if they issued international debt during the financial crisis.

Previous studies examining the effect of the 2007–2009 financial crisis on corporations show that the investment policies of the U.S. incorporated firms were affected during the crisis (see, e.g., Duchin, Ozbas, and Sensoy 2010). Supplementing this evidence, I document that the financing channel of internationally diversified firms was less impaired by the financial shock, and partly as a result, the U.S. investment policy of multinationals was less affected than that of domestic firms. The analysis on the domestic investment policies suggests that financial flexibility indeed was valuable to multinational firms during the financial crisis.

5. Robustness Tests

5.1 Selection bias on bank loan and corporate bond samples

The sample of bank loan and corporate bonds does not include firms that have not received bank loans or issued corporate bonds. One potential concern is that the results could be driven by unobservable firm characteristics that

[30] Within a subset of multinational firms that have subsidiaries in less affected countries, about 75% of firms have subsidiaries in Canada, 45% in Australia, 39% in Singapore, and 32% in Spain or in Brazil. I use alternative definitions for *Less affected sub* considering the presence of foreign subsidiaries in less affected countries except Canada and European countries. The results do not change quantitatively, which implies that the effect of operating in countries that were not largely affected by financial crisis periods is not primarily driven by a few specific countries (e.g., Canada).

Table 8
U.S. investment of multinationals during the financial crisis: Foreign debt issuance and location of foreign subsidiaries

	(1)	(2)	(3)	(4)
	OLS			
Dependent variable:	U.S. expansion announcement			
Early crisis	−0.036***	0.002	0.015	0.024
	(0.014)	(0.033)	(0.010)	(0.024)
Late crisis	−0.080***	−0.100**	−0.022*	0.001
	(0.018)	(0.045)	(0.013)	(0.032)
Post-crisis	−0.022	−0.064	0.004	−0.022
	(0.017)	(0.042)	(0.011)	(0.027)
Less affected sub	−0.052**	−0.029		
	(0.023)	(0.053)		
Early crisis x Less affected sub	0.072***	0.038		
	(0.014)	(0.033)		
Late crisis x Less affected sub	0.079***	0.140***		
	(0.019)	(0.046)		
Post-crisis x Less affected sub	0.035**	0.051		
	(0.017)	(0.041)		
Issue foreign debt			0.039***	0.034*
			(0.009)	(0.020)
Early crisis x Issue foreign debt			0.077***	0.106*
			(0.025)	(0.059)
Late crisis x Issue foreign debt			0.086**	0.099
			(0.042)	(0.091)
Post-crisis x Issue foreign debt			0.029	−0.061
			(0.029)	(0.073)
Controls	Firm-level	U.S.-level	Firm-level	U.S.-level
Firm FE	Yes	Yes	Yes	Yes
Observations	55,914	9,873	55,914	9,873
Adj R-squared	0.205	0.213	0.206	0.213

This table presents the effect of the 2007–2009 financial crisis on the domestic investment of multinational firms depending on their foreign subsidiary locations and international debt financing during the crisis period. The sample is restricted to the panel of firm-quarter observations of multinationals only from 2000 to 2010. The table reports estimates of OLS regressions in which the dependent variable is the U.S. expansion announcement indicator variable that takes a value of one if a firm makes an announcement of either an acquisition of U.S. firms or U.S. expansion in a given quarter. All regressions include firm characteristics as control variables as is done in Columns 1 and 2 of panel B of Table 7, but the coefficients of control variables are not reported to save space. All regressions include firm fixed effects. *Early crisis* is defined as calendar quarters between 2007Q3 and 2008Q3, *Late crisis* as 2008Q4-2009Q1, and *Post-crisis* as 2009Q2-2010Q1. *Less affected sub* is defined as one if a multinational firm has at least one subsidiary in the countries in which the change in the ratio of private credit to GDP between 2007 and 2009 is above the top tercile among 53 countries. *Issue foreign debt* is defined as one if a multinational firm issues a bank loan from a foreign lead lender or an international bond in a quarter. Table A1 (see the appendix) defines the variables. The standard errors with clustering at the firm level are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

influence the firm's ability to raise capital and are correlated with the variable denoting multinationality. For example, only higher quality firms could receive the loans in bad times. If those firms tend to be multinationals and if they borrow from foreign lenders, the sample of loan issuances would be composed of disproportionally more loans issued by multinationals from foreign lenders.

To evaluate the extent to which this potential selection bias influences the results, I estimate Heckman (1979) sample selection models for the baseline regressions in Table 3. I collapse the bank loan and corporate bond issuance data into firm-quarter observations and merge this data set with the universe of Compustat firms on a quarterly basis to include all firms that do not issue

Table 9
Heckman selection model controlling for the probability of debt issuance

	(1)	(2)	(3)	(4)	(5)
	Selection	Outcome		Selection	Outcome
Dependent variable:	Issue loan	Have foreign lender	Have foreign lead	Issue bond	International bond
LT debt maturing 1 yr	0.185***			0.066	
	(0.030)			(0.047)	
GlobalDiv	0.035**	0.117***	0.030**	−0.015	0.051***
	(0.015)	(0.021)	(0.013)	(0.020)	(0.018)
Log(sales)	0.125***	0.144***	−0.017	0.195***	0.098
	(0.006)	(0.023)	(0.015)	(0.007)	(0.076)
Sales frowth	0.070***	0.072***	0.022	0.172***	0.106
	(0.019)	(0.026)	(0.015)	(0.024)	(0.067)
Cash flows	−0.443*	−0.143	−0.178	−3.407***	−0.965
	(0.240)	(0.324)	(0.199)	(0.310)	(1.282)
Market-to-book	0.011	0.019*	−0.000	0.053***	0.000
	(0.008)	(0.010)	(0.007)	(0.010)	(0.022)
Leverage	0.350***	0.306***	0.052	0.906***	0.423
	(0.037)	(0.072)	(0.047)	(0.048)	(0.360)
Cash	−1.348***	−0.266	0.519***	−0.218***	−0.037
	(0.066)	(0.264)	(0.171)	(0.080)	(0.103)
STD(cash flows)	−0.076	−1.585***	−0.397	0.476	0.108
	(0.414)	(0.565)	(0.371)	(0.546)	(0.457)
Lambda		0.051	−0.385***		0.264
		(0.218)	(0.142)		(0.480)
Rating and quarter FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Observations	P86,506	7,576	7,576	86,506	4,126
Censored observations	78,930			82,380	
Pseudo/Adj R-squared		0.300	0.116		0.333

This table presents estimates of the international corporate diversification on foreign lender participation in bank loans and international bond issuance in corporate bonds using Heckman selection models. The bank loan and corporate bond issue data are collapsed into firm-quarter observations and are merged to the universe of Compustat firms at the quarterly basis to include all firms that do not issue loans or bonds. Columns (1) and (4) report the results from probit models of selection equations that estimate the probability of issuing bank loans or the probability of issuing bonds. The dependent variable is equal to one if a firm issues a loan (or a bond) in a given quarter. An instrumental variable predicting debt issues in the selection equation is the proportion of long-term debt maturing within a year in the previous quarter. The estimations from the probit models of outcome equations that adjust for selection bias are reported in Columns 2, 3, and 5. In the outcome equations, the dependent variables are foreign lender participation indicators in Columns 2 and 3 and the indicator for the international bond issuance in Column 5. All regressions include *GlobalDiv*, firm characteristics, rating fixed effects, industry fixed effects, and quarter fixed effects. The standard errors adjusted for heteroscedasticity and firm-level clustering are in parentheses. ***, **, and * indicate p-values of 1%, 5%, and 10%, respectively.

loans or bonds. A selection equation estimates the probability of issuing bank loans or the probability of issuing bonds each quarter using a probit model. The procedure requires an instrument in the selection equation that is correlated with debt issue decisions but has no effect on foreign lender participation and international bond issuance. As suggested by Almeida et al. (2011), I use the ratio of long-term debt maturing within one year to total long-term debt as an instrument variable. In outcome equations, I use a bivariate probit model adjusted for selection bias for a binary dependent variable, and I use Heckman selection model for a continuous dependent variable.

Table 9 displays the results. In the selection regression reported in Column 1, the instrument variable, the proportion of long-term debt maturing within a

Table 10
Alternative measures of the financial crisis

	(1)	(2)	(3)	(4)	(5)	(6)
	Bank loans			Corporate bonds		
	OLS			Probit		
Dependent variable:	Foreign lead lender shares			International bond		
GlobalDiv	−0.003	0.005	0.007	−0.052	0.065**	0.069**
	(0.010)	(0.006)	(0.006)	(0.068)	(0.032)	(0.030)
VIX Index	−0.000			0.002		
	(0.000)			(0.002)		
GlobalDiv x VIX Index	0.001**			0.006**		
	(0.000)			(0.003)		
TED spread		−0.016***			0.065*	
		(0.005)			(0.038)	
GlobalDiv x TED spread		0.024***			0.029	
		(0.009)			(0.043)	
CP spread			−0.025***			0.011
			(0.007)			(0.064)
GlobalDiv x CP spread			0.034**			0.023
			(0.014)			(0.071)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,639	11,639	11,639	3,406	3,406	3,406
Pseudo/Adj R-squared	0.0823	0.0827	0.0826	0.379	0.374	0.369

This table presents the main baseline regression results in panel A of Table 5 with alternative measures of the financial crisis indicators. The table reports estimates of OLS regressions in which the dependent variable is foreign lender shares, using a sample of bank loans in Columns 1 to 3, and marginal effects of probit regressions in which the dependent variable is international bond issuance indicator in Columns 4 to 6. Instead of the financial crisis quarter dummies, the regressions include the continuous variables that capture the intensity of the crisis and the interaction terms of *GlobalDiv* with each financial crisis intensity measure: *VIX index* (Chicago Board Options Exchange Volatility Index), *TED spread* (the three-month LIBOR over Treasury-bill yields), and *CP spread* (the three-month commercial paper rate over Treasury-bill yields). All regressions include firm, loan, and bond characteristics as control variables as is done in panel Al of Table 5, but the coefficients of control variables are not reported to save space. Table A1 (see the appendix) defines the variables. The standard errors with clustering at the firm level are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

year, strongly predicts the firm's decision to issue a bank loan. After correcting for the sample selection, the global diversification indicator variable still remains statistically significant in explaining the foreign lender participation in Columns 2 and 3. Next, I turn to selection models for corporate bond issuance in Table 9. In the selection equation, however, the proportion of long-term debt maturing within a year does not have predictive power. The results in Column 5 suggest that the positive effect of the global diversification on the probability of issuing bonds in international markets is robust.

5.2 An alternative definition for the financial crisis

As the main variables for the financial crisis, I use the crisis indicator variables that capture the timing of financial crisis at the quarterly basis. As a robustness check, I also confirm the main results with the alternative measures that estimate the extent to which a firm might have difficulties in receiving external financing in the U.S. markets. Instead of the financial crisis indicators, I use three continuous variables: *VIX index*, *TED spread*, and *CP spread*. It has been documented that the spreads on short-term financing instruments increased sharply and reached historically high levels during the financial crisis (e.g., Almeida et al. 2011).

Table 10 reports the results. In Columns 1 to 3, the results from the OLS regressions suggest that the positive impact of being multinational on foreign lead lender shares becomes stronger when the VIX index, the TED spread, and CP spread are high. On the other hands, the results on corporate bond issuances are relatively weaker. In Columns 4 to 6, I find that the interaction term of *GlobalDiv* with the VIX index is only statistically significant and positive. These results confirm that my previous findings are robust to alternative measures for the crisis period, which estimate the intensity of the shocks in external financing markets.

5.3 Reverse causality

It is possible that firms expand operations to the specific countries in which they have already attained access to capital. To examine this concern, I use a subsample of firms that become multinational during the sample period and issue at least one bank loan before and after the diversification decision. I consider whether the foreign lender participation and the propensity to issue international bond increase after the firms become multinational. In unreported results, I find that the propensity to have a foreign lender increases after the firms become internationally diversified. I find a similar result using corporate bond issuance. This test confirms the finding that foreign lender participation is due to multinationality.

6. Conclusion

This paper studies how international corporate diversification affects firm financing policies, especially their access to foreign funding sources in bank loans and corporate bonds. Using a sample of U.S. multinationals and domestic firms from 2000 to 2010, I have documented that firms that had foreign operations make more use of funding in foreign countries than did firms that had purely domestic operations when firms issue bank loans. The location of multinationals' operations strongly predicts the nationality of foreign lenders. The higher foreign lender participation in bank loans to multinationals is mostly attributable to foreign lead lenders from the same region and from countries in which multinationals have foreign subsidiaries. In addition, conditional on corporate bond issuance, bonds issued by multinationals are more likely to be placed in countries outside the United States.

The access to funding in foreign countries by multinationals leads to greater financial flexibility when a capital market is disrupted in their home country since they can exploit foreign funding sources more easily than purely domestic firms. Using the 2007–2009 financial crisis as a capital supply shock, I have shown that multinationals were better able to borrow more from foreign lead lenders than were domestic firms at the peak of the financial crisis (the two quarters following the Lehman Brothers bankruptcy) relative to the period outside the financial crisis. This effect is mostly driven by the increased participation of lenders from countries in which multinationals have foreign subsidiaries and in which bank lending was less affected. Moreover, multinationals used international bond markets more actively than did purely domestic firms in the two quarters after September 2008. Partially as a consequence, the U.S. investments of multinationals were less adversely affected during the crisis than those of domestic firms.

Overall, the empirical evidence in this paper is consistent with the hypothesis that having an operational presence in foreign countries provides U.S. multinational firms with greater financial flexibility than otherwise similar U.S. domestic firms. One financial advantage of being multinational is that it allows firms to access international debt markets at lower costs during the times of capital market disruptions in the home country. This study suggests several financial aspects to multinational firms go beyond merely the ability to produce and sell products in different markets.

Appendix

Table A1
Description of variables

Variable	Description
A. Foreign operation involvement	
GlobalDiv	An indicator variable equal to one if a firm's foreign pretax income (*pifo*) or foreign income tax (*txfo*) is not missing in at least one year over the previous three years and it has at least one subsidiary outside the U.S. subsidiaries in tax-haven countries are not counted (*Source*: Compustat Annual, Capital IQ, OECD)
% foreign income	The proportion of absolute value of foreign pretax income (*pifo*) to the sum of absolute value of foreign pretax income (*pifo*) and absolute value of domestic pretax income income (*pidom*) (*Source*: Compustat Annual)
% foreign sales	The proportion of foreign sales to total sales (*Source*: Compustat Geographic Segment)
HHI(sales)	One minus Herfindahl index of sales concentration $1-\sum_{i=1}^{N}(Sales_i/Totalsales)^2$ in which $Sales_i$ is sales of geographic segment i and N is the total number of geographic segments. (*Source*: Compustat Geographic Segment)
Less affected sub	An indicator variable equal to one if a multinational firm has at least one subsidiary in the countries in which the change in the ratio of private credit to GDP between 2007 and 2009 is above the top tercile (*Source*: Capital IQ, IMF's International Financial Statistics)
Sub high private credit	An indicator variable equal to one if a multinational firm has at least one subsidiary in the countries in which the private credit by deposit money banks and other financial institutions to GDP ratio is in the top tercile in a given year (*Source*: Capital IQ, IMF's International Financial Statistics)
Sub strong creditor right	An indicator variable equal to one if a multinational firm has at least one subsidiary in the countries in which the creditor rights index as defined by Djankov, McLiesh, and Shleifer (2007) is in the top tercile (*Source*: Capital IQ)
Sub distant	An indicator variable equal to one if a multinational firm has at least one subsidiary in the countries in which the geographic proximity is in the bottom tercile. Geographic proximity is measured as the negative of the great circle distance between the capitals of countries (*Source*: Capital IQ)
Sub unfamiliar	An indicator variable equal to one if a multinational firm has at least one subsidiary in the countries in which the maximum of bilateral import and export with the United States is in the bottom tercile in a given year (*Source*: Capital IQ, UN commodity trade database)

(*continued*)

Table A1
Continued

Variable	Description
B. Firm characteristics (*Source:* Compustat Quarterly and Annual)	
Total assets	Total assets (*atq*) in 2005 US dollars
Market cap	The market value of common equity (*cshoq*prccq*) in 2005 US dollars
Log(sales)	Natural log of annual sales (*saleq*) in 2005 US dollars
Leverage	The sum of long-term debt (*dlttq*) and debt in current liabilities (*dlcq*) divided by total assets (*atq*)
Sales growth	Change in sales (*saleq*) divided by the one-year lagged sales. To adjust seasonality, the change in sales is calculated by subtracting the sales at the same quarter of previous fiscal year
Cash flows	Operating income before depreciation (*oibdq*) divided by total assets (*atq*)
Cash	Cash and marketable securities (*cheq*) divided by total assets (*atq*)
Market to book	The ratio of market value of assets to book value of assets (*atq*) in which market value of assets is calculated as book value of assets (*atq*) minus book value of common equity (*ceqq*) plus the market value of common equity (*cshoq*prccq*)
STD(cash flows)	Standard deviation of cash flows to total assets in previous 20 quarters
Capex	Capital expenditures (*capxy*) divided by lagged total assets (*atq*). Since *capxy* is a year-to-date basis variable, the quarterly value is calculated by subtracting the lagged variable from current one except the first quarter of a fiscal year.
U.S. capital expenditure	Annual U.S. segment capital expenditures divided by lagged total assets (*at*) for multinational firms and capital expenditures (*capx*) divided by lagged total assets (*at*) for domestic firms (*Source*: Compustat Geographic Segment)
U.S. expansion announcement	An indicator variable of quarters when a firm makes either a U.S. acquisition or U.S. expansion announcement. U.S. acquisitions include completed transactions of purchase of partial/entire assets or stakes on U.S. target firms in SDC. U.S. expansions include “Seeking Acquisitions/Investments” and “Business Expansion” events that are related to U.S. operations from the “Key Development” database from Capital IQ (*Source*: SDC, Capital IQ)
S&P credit rating	S&P long-term public bond rating (*splticrm*) are coded into eight categories as AAA, AA, A, BBB, BB, B, CCC+, and below and unrated
C. Loan characteristics (*Source:* DealScan)	
Have foreign lenders	An indicator variable equal to one if the loan syndicate includes at least one foreign lender. A foreign lender is defined as a lender incorporated outside of the United States, excluding a foreign branch of the U.S. banks or a U.S. branch of foreign banks
Foreign lender share	The proportion of loan amount originated by foreign lenders to the total facility amount. For loans with missing share variable, I assume each lender in the facility deal takes an equal share
Have foreign lead lenders	An indicator variable equal to one if any lead lender in the loan syndicate is a foreign lender. A foreign lender is defined as a lender incorporated outside of the United States, excluding a foreign branch of the U.S. banks or a U.S. branch of foreign banks. A lender is defined as a lead lender if “Lead Arranger Credit” is equal to “Yes” in DealScan, if a lender is identified as “Agent,” “Administrative Agent,” “Arranger,” and “Lead Bank,” or if the loan is a sole-lender loan, following Bharath, Dahiya, Saunders, and Srinivasan (2011)
Foreign lead lender share	The proportion of loan amount originated by foreign lead lenders to the total facility amount
Log(facility amount)	Natural log of the loan facility size
Log(number of lenders)	Natural log of the number of lenders in the loan facility syndicate
Log(maturity)	Natural log of the maturity of the loan facility in months
Secured	An indicator variable equal to one if the loan facility is secured and equal to zero otherwise. If the variable is missing, it is set to zero
Missing secured	An indicator variable equal to one if the information on whether the loan is secured or not is not available from DealScan

(*continued*)

Table A1
Continued

Variable	Description
D. Bond characteristics (*Source:* SDC)	
International bond	An indicator variable equal to one if a bond is placed in an exchange outside the United States, if it is a Euro bond, or if it is a global bond. International bonds include the following types of bonds: Global Notes, Global Bonds, Global MTNs, Global FRNs, Global Debts, Global MTN Program, Euro CP Program, and Euro MTN Program, as defined in SDC
Log(proceed amount)	Natural log of proceed amount in US dollars
Log(maturity)	Natural log of bond maturity in months
Secured	An indicator variable equal to one if the bond is secured
Private debt	An indicator variable equal to one if the bond is privately placed
Callable	An indicator variable equal to one if the bond is callable
E. Macroeconomic variables	
Pre-crisis	An indicator variable of quarters from January 2000 to June 2007
Early crisis	An indicator variable of quarters from July 2007 to August 2008
Late crisis	An indicator variable of quarters from September 2008 to March 2009
Post-crisis	An indicator variable of quarters from April 2009 to March 2010
VIX index	Chicago Board Options Exchange Volatility Index (*Source*: Chicago Board Options Exchange)
TED spread	The difference between the three-month LIBOR and the yield on three-month Treasury bills (*Source*: Federal Reserve Bank of St. Louis Economic Data)
CP spread	The difference between the three-month commercial paper and the yield on three-month Treasury bills (*Source*: Federal Reserve Bank of St. Louis Economic Data)

Table A2
Propensity-score-matching estimation results

	(1)	(2)	(3)	(4)
	Bank loans		Corporate bonds	
	Pre-match	Post-match	Pre-match	Post-match
Log(sales)	0.111***	0.024	−0.003	0.037
	(0.013)	(0.017)	(0.020)	(0.035)
Leverage	−0.156***	0.033	−0.337***	0.099
	(0.055)	(0.072)	(0.110)	(0.197)
Sales growth	−0.07561***	−0.008	−0.146***	−0.037
	(0.024)	(0.033)	(0.041)	(0.093)
Cash flows	−0.46164	−0.477	−1.293**	−0.778
	(0.281)	(0.434)	(0.504)	(0.864)
Cash	0.41363***	−0.181	0.987***	0.273
	(0.103)	(0.126)	(0.286)	(0.449)
Market-to-book	0.025**	0.018	0.034	0.021
	(0.010)	(0.012)	(0.024)	(0.047)
STD(cash flows)	−2.466***	−0.607	−4.588***	−1.005
	(0.652)	(0.925)	(1.330)	(2.194)
Log(facility/proceed amount)	0.008	−0.010	0.013	−0.055*
	(0.009)	(0.013)	(0.018)	(0.029)
Log(maturity)	0.020	−0.006	−0.057*	−0.128***
	(0.012)	(0.016)	(0.031)	(0.045)
Secured	−0.030**	−0.005	−0.048	0.004
	(0.013)	(0.019)	(0.053)	(0.089)
Missing secured	0.008	−0.002		
	(0.024)	(0.034)		
Log(number of lenders)	0.025	−0.026		
	(0.022)	(0.032)		
Private debt			−0.057	0.012
			(0.036)	(0.055)
Callable			0.059*	−0.067
			(0.032)	(0.056)
Rating and industry FE	Yes	Yes	Yes	Yes
Loan purpose and loan type FE	Yes	Yes	No	No
Domestic firms	5,184	5,736	1,197	1,473
Domestic firms (unique obs)	5,184	2,386	1,197	519
Multinational firms	6,455	5,736	2,209	1,473
Observations	11,639	11,472	3,406	2,946
Pseudo R-squared	0.212	0.00783	0.170	0.0345

The table presents the results of a probit regression with the indicator of multinational as the dependent variable using the bank loan sample in Columns 1 and 2 and using the corporate bond sample in Columns 3 and 4. In *Pre-match* model, the entire sample of firms is used and in *Post-match* model, only the banks loans or corporate bonds that are matched using the propensity score from the pre-match model are used. The standard errors with firm-level clustering are in parentheses. ***, **, and * indicate p-values of 1%, 5%, and 10%, respectively.

Table A3
Access to foreign bank loans: Location of foreign subsidiaries by country

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Have lender from:									
Dependent variable:	Canada	France	Japan	Germany	Netherlands	Switzerland	Hong Kong	Italy	Belgium	Australia
Sub in Canada	**0.045***	−0.027	0.028	−0.011	−0.023	0.007	0.001	0.001	0.002	−0.002
	(0.023)	(0.023)	(0.019)	(0.017)	(0.019)	(0.012)	(0.007)	(0.001)	(0.003)	(0.003)
Sub in France	0.038	**0.046**	−0.019	0.006	−0.011	0.001	0.002	0.000	−0.005	0.001
	(0.031)	**(0.029)**	(0.025)	(0.019)	(0.020)	(0.015)	(0.008)	(0.001)	(0.004)	(0.003)
Sub in Japan	−0.025	0.033	**0.081*****	−0.024	0.001	0.009	−0.008	0.001	−0.004	−0.000
	(0.026)	(0.027)	**(0.025)**	(0.016)	(0.018)	(0.012)	(0.006)	(0.001)	(0.004)	(0.002)
Sub in Germany	−0.025	−0.008	0.007	**0.033***	0.005	0.012	−0.007	0.000	0.004	0.003
	(0.028)	(0.027)	(0.024)	**(0.018)**	(0.021)	(0.014)	(0.008)	(0.001)	(0.005)	(0.002)
Sub in Netherlands	−0.014	0.010	0.017	−0.001	**0.049*****	0.001	0.005	−0.000	−0.003	0.003
	(0.025)	(0.025)	(0.020)	(0.017)	**(0.018)**	(0.013)	(0.007)	(0.001)	(0.004)	(0.002)
Sub in Switzerland	−0.033	0.029	0.019	−0.028*	0.030	**0.014**	−0.011*	0.001	−0.001	0.001
	(0.027)	(0.029)	(0.026)	(0.015)	(0.021)	**(0.015)**	(0.006)	(0.001)	(0.004)	(0.003)
Sub in Hong Kong	−0.020	−0.050**	0.006	0.018	0.004	−0.012	**0.000**	0.001	−0.006*	0.004
	(0.025)	(0.024)	(0.023)	(0.018)	(0.019)	(0.012)	**(0.006)**	(0.001)	(0.003)	(0.003)
Sub in Italy	−0.003	−0.002	0.024	0.051**	−0.022	0.001	0.022**	**0.003**	0.001	−0.006**
	(0.029)	(0.028)	(0.026)	(0.020)	(0.020)	(0.014)	(0.010)	**(0.002)**	(0.004)	(0.003)
Sub in Belgium	0.008	−0.018	−0.019	−0.029*	0.019	−0.008	0.001	−0.001*	**0.031*****	−0.005**
	(0.028)	(0.029)	(0.023)	(0.017)	(0.021)	(0.014)	(0.008)	(0.001)	**(0.012)**	(0.002)
Sub in Australia	0.019	0.012	−0.009	−0.001	0.012	−0.004	0.012*	−0.001	0.003	**0.010****
	(0.023)	(0.024)	(0.021)	(0.016)	(0.019)	(0.012)	(0.006)	(0.000)	(0.004)	**(0.005)**
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,455	6,455	6,455	6,455	6,455	6,434	6,455	6,012	6,175	6,220
Pseudo R-squared	0.297	0.354	0.445	0.329	0.288	0.188	0.279	0.518	0.301	0.344
% MNCs that have lender from each country	30.74	28.92	30.69	22.56	19.58	8.30	11.88	8.43	6.07	4.55

This table presents results from regressions estimating the effect of international corporate diversification on access to foreign capital markets depending on the location of multinationals' foreign subsidiaries by country. The sample is restricted to the loans to multinationals. I report marginal effects from probit models estimating the probability of having a lender from each country. I consider the top ten countries (Canada, France, Japan, the United Kingdom, Germany, Netherlands, Switzerland, Hong Kong, Italy, and Belgium), from which foreign lenders are the most active in U.S. bank loan markets. The regressions include firm and loan characteristics with rating, industry, and quarter fixed effects, as is done in Column 2 of Table 3. The coefficients of control variables are not reported to save space. Table A1 (see the appendix) defines all variables. The standard errors with clustering at the firm level are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table A4
The 2007–2009 financial crisis and foreign lender participation by credit rating

	(1)	(2)		(3)	(4)	
Dependent variable:	Have foreign lead lender			Foreign lead lender shares		
Subsample:	Rated	Unrated	Diff. p-value	Rated	Unrated	Diff. p-value
GlobalDiv	0.063***	0.010		0.012*	0.009	
	(0.022)	(0.008)		(0.006)	(0.006)	
Early crisis	0.027	0.993***		−0.040*	0.032	
	(0.094)	(0.003)		(0.023)	(0.021)	
Early crisis x GlobalDiv	0.034	0.013	.896	0.015	0.033**	.410
	(0.061)	(0.023)		(0.014)	(0.016)	
Late crisis	0.146	0.189		0.021	0.027*	
	(0.172)	(0.321)		(0.044)	(0.016)	
Late crisis x GlobalDiv	0.036	0.971***	.000	0.067	0.081*	.735
	(0.121)	(0.004)		(0.054)	(0.042)	
Controls	Yes	Yes		Yes	Yes	
Observations	6,337	5,302		6,337	5,302	
Pseudo/Adj R-squared	0.0958	0.187		0.147	0.0819	
Mean interaction effect for						
Early crisis x GlobalDiv	0.036	−0.002				
	[0.058]	[0.023]				
Late crisis x GlobalDiv	0.073	0.123				
	[0.517]	[1.022]				

The table presents estimates of regressions from panel A of Table 5 for subsamples based on the existence of S&P long-term public bond rating. Columns (1) and (3) examine a subsample of firms that have S&P long-term public bond rating when receiving bank loans, and Columns 2 to 4 examine a subsample of firms that do not have rating. The regression specifications follow Columns 2 and 4 in panel A of Table 5. I report p-values for the difference in coefficients of each variable from the joint estimation of two probit and OLS models. The coefficients on control variables are not reported to save space. On the bottom of the table, I report the mean interaction effects and z-statistics corrected by the methodology of Norton, Wang, and Ai (2004). The standard errors with firm-level clustering are in parentheses. ***, **, and * indicate p-values of 1%, 5%, and 10%, respectively.

Table A5
International corporate diversification and U.S. investment policy during the financial crisis: Propensity-score-matching estimation

	U.S. capital expenditure									
	A. Firm-level variable match					*B. U.S.-level variable match*				
	Paired obs.	Multinational	Domestic	Difference (SE)		Paired obs.	Multinational	Domestic	Difference (SE)	
(1) 2000–2005	708	0.0439	0.0748	−0.0309	(0.004)***	481	0.0469	0.0820	−0.0351	(0.005)***
(2) Crisis2008	143	0.0491	0.0898	−0.0407	(0.011)***	119	0.0491	0.0827	−0.0336	(0.011)***
(3) Crisis2009	137	0.0344	0.0617	−0.0273	(0.007)***	110	0.0339	0.0490	−0.0151	(0.007)**
(4) Crisis2010	107	0.0422	0.0546	−0.0124	(0.009)	79	0.0464	0.0692	−0.0228	(0.013)*
(2) - (1)				−0.0098	(0.010)				0.0015	(0.011)
(3) - (1)				0.0036	(0.010)				0.0200	(0.011)*
(4) - (1)				0.0185	(0.011)*				0.0123	(0.013)
	U.S. expansion announcement									
	C. Firm-level variable match					*D. U.S.-level variable match*				
	Paired obs.	Multinational	Domestic	Difference (SE)		Paired obs.	Multinational	Domestic	Difference (SE)	
(1) Pre-crisis	33,457	0.1634	0.1706	−0.0072	(0.003)**	6,506	0.1386	0.1543	−0.0157	(0.006)**
(2) Early crisis	6,626	0.3156	0.2996	0.0160	(0.008)**	1,408	0.2855	0.2202	0.0653	(0.016)***
(3) Late crisis	2,558	0.2885	0.2432	0.0453	(0.012)***	2,558	0.2845	0.1650	0.1195	(0.024)***
(4) Post-crisis	4,944	0.3081	0.2530	0.0551	(0.009)***	4,944	0.2717	0.1946	0.0771	(0.017)***
(2) - (1)				0.0232	(0.007)***				0.0810	(0.015)***
(3) - (1)				0.0525	(0.011)***				0.1352	(0.022)***
(4) - (1)				0.0623	(0.008)***				0.0928	(0.016)***

This table reports the average differences of U.S. capital expenditures and U.S. expansion announcements between the multinationals and the matched domestic firms during the financial crisis using the propensity-score score-matching sample. Propensity scores are obtained from the probit model predicting whether the firm is multinational. Control variables in the probit model include *Log(sales), Sales growth, Cash flows*, and *Market-to-book* as is done in Column 1 of panel A of Table 7. The matching firm-level variables in the probit model are estimated as of the end of fiscal year of 2006 in panels A and B (as of 2006Q3 in panels C and D). They are estimated at the firm-level in panels A and C and at the U.S.-level in panels B and D. Each multinational firm (treated group) is matched to a domestic firm (control group) within the same S&P credit rating category in the same year (or quarter), using propensity score estimation, with replacement. I apply the nearest neighbor-matching estimator like in Becker and Ichino (2002) of 0.1 caliper and imposing the common support condition. Panels A and B report the differences in annual U.S. capital expenditures and panels C and D report the differences in quarterly U.S. expansion announcements by subperiod. Each subperiod is defined as *Pre-crisis* (2000Q1-2007Q2), *Early crisis* before the collapse of Lehman (2007Q3-2008Q3), *Late crisis* after the collapse of Lehman (2008Q4-2009Q1), and *Late crisis* (2009Q2-2010Q1). Table A1 (see the appendix) defines all variables. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

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